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Modular ML-Based Framework for Blockchain Fraud Detection Across Layer-2 and Cross-Chain Ecosystems

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1. Context & Problem Statement

This section introduces the evolving challenges of financial crime in blockchain ecosystems, explains why conventional systems are insufficient, and motivates the development of a new detection framework to address these issues.

1.1. The Changing Nature of Blockchain-Based Financial Crime

Blockchain's foundational attributes like full transparency, deterministic immutability, and permissionless composability, were initially celebrated as breakthroughs for trustless finance. Over time, these same properties have been leveraged for increasingly sophisticated illicit activity.

Illicit cryptocurrency activity reached unprecedented levels in 2024, driven not only by more frequent exploits but also by advanced laundering strategies involving stablecoins, cross-chain bridges, and state-sponsored actors. These developments show that the tools designed to democratise finance have also created powerful infrastructure for obfuscation.

Two factors stand out:

- **Transparency and mempool visibility**: Public ledgers and mempools allow adversaries to observe pending transactions and adapt their tactics in real time. Stolen funds can be moved into mixers or cross-chain bridges moments before an on-chain alert is triggered, concealing the true source of illicit funds.
- **Irreversible immutability**: Once a transaction is confirmed, there is no mechanism to reverse it, making on-chain settlement a preferred tool for ransomware operators and exploiters.

The rise of Ethereum Layer-2 networks and cross-chain protocols has further accelerated laundering efficiency. High-throughput, low-cost transaction environments that are beneficial for legitimate use, also provide criminals with faster, more discreet routes that fragment the audit trail. For example, Ethereum Layer 1 processes millions of daily transactions, but Layer-2 networks collectively handle several times that volume, dispersing transaction records across multiple layers and storage systems.

Case studies show how these environments are exploited:

- Large-scale mixer inflows before sanctions or shutdowns.
- Layer-2 and cross-chain laundering routes becoming standard in DeFi-related exploits.
- State actors chaining bridge-hops with synthetic asset minting to break transaction lineage.

The compliance challenge lies in **distinguishing between legitimate privacy-preserving actions** (e.g., consolidating funds through a mixer) and laundering schemes that employ similar tools to disguise illicit proceeds.

1.2. Why Traditional AML Systems Fall Short

Traditional Anti-Money Laundering (AML) frameworks were built for centralised financial systems, where institutions have complete visibility over identity, account activity, and transaction flows. These systems assume:

1. **Identity consistency** – the same entity can be tracked across all activities.
2. **Continuous value flow** – funds can be traced from origin to destination.
3. **Static perimeters** – transactions occur within a monitored environment.

In decentralised finance, none of these assumptions hold. Pseudonymity, composability, and cross-chain mobility enable laundering behaviours that bypass traditional thresholds and monitoring logic. Examples include:

- **Transaction fragmentation**: High-value transfers split into smaller amounts and routed through short-lived wallets.
- **Flash loans**: Instant, collateral-free borrowing enabling value movement within a single block.
- **Mixers and cross-chain bridges**: Tools that break transaction continuity across incompatible ledgers.
- **Layer-2 logging constraints**: Off-chain or compressed data that reduce real-time visibility.
- **Synthetic mint/burn cycles**: Replication of value across chains without retaining historical linkages.

These behaviours lead to high false positives flagging legitimate privacy users and high false negatives, where complex laundering patterns evade detection entirely.

1.3. Gaps in Existing Blockchain Analytics

While blockchain analytics platforms have enabled powerful tools for tracing illicit activity on-chain, most operate reactively, offering limited capacity for real-time or proactive fraud detection. This section outlines the core shortcomings of current systems and motivates the need for a new approach.

Blockchain analytics refers to specialized tools and platforms, such as Chainalysis or Elliptic, that analyse on-chain data to identify suspicious patterns and trace illicit flows. These platforms extract insights from public transaction data to help track illicit fund movements across wallets, exchanges, and smart contracts.

While these tools provide valuable insights, they often focus on historical tracking rather than proactive detection. Blockchain-native analytics platforms can reconstruct laundering trails after the fact, but they struggle in fast-moving, multi-chain contexts.

Common limitations include:

- **Opaqueness**: Risk models and decision logic are often proprietary and cannot be independently verified.
- **Post-event detection**: Alerts typically occur after the laundering process is complete.
- **Limited behavioural differentiation**: Risk scores rarely distinguish between suspicious activity and privacy-preserving actions.

The core challenge remains **moving from reactive investigation to proactive detection** and doing so without misclassifying legitimate users.

1.4. Thesis Motivation

The accelerating complexity of blockchain ecosystems driven by cross-chain protocols, Layer-2 networks, and privacy tools, has exposed the limits of existing fraud detection systems. This section presents the motivation behind developing a new modular and interpretable detection framework.

The rapid evolution of blockchain-based financial crime has outpaced the capabilities of traditional Anti-Money Laundering (AML) systems and many blockchain analytics platforms. The combination of cross-chain laundering, Layer-2 adoption, and privacy tool usage has created detection challenges that require new, adaptive approaches.

This thesis proposes a **modular, interpretable, and machine learning–driven fraud detection framework** designed for Ethereum and its Layer-2 ecosystems. The framework is designed to:

1. Improve detection accuracy through a hybrid supervised–unsupervised pipeline.
2. Differentiate behaviours to avoid misclassifying legitimate privacy use as fraud.
3. Provide explainability suitable for compliance, enabling alerts to be traced back to clear behavioural triggers.

Key Attributes of the Proposed Framework

Attribute	Description
Transparent	Open-source models with explainable decisions (e.g., reason codes, SHAP)
Real-time Aware	Simulated or actual use on live transaction streams
Behavioural	Uses transaction patterns, not just blacklists
Scalable	Modular features (L0–L3) and layer-wise scripts
Cross-chain aware	Tracks bridge flows, asset hops, synthetic minting
Stablecoin analytics	Detects suspicious flows using USDT/USDC, including mint/burn laundering
Sanctions-linked	Flags suspicious USDT/USDC mint–bridge–burn flows
Adaptive obfuscation	Reacts to emerging tools like YoMix, replacing mixers
State actors signal aware	Tracks tactics of actors like Lazarus (timing, asset swaps, routing logic)

By combining behavioural modelling, real-time adaptability, and regulatory-aligned explainability, the framework aims to deliver a practical, reproducible, and scalable approach to fraud detection in the increasingly complex Ethereum and Layer-2 landscape.

2. Literature Review and Research Alignment

This section reviews existing research on blockchain fraud detection, organized by key drivers aligned with the thesis's focus areas, and identifies gaps that inform the strategic objectives and design of the proposed framework.

2.1. Overview

Research on blockchain-based financial crime has evolved alongside the growth of decentralised finance (DeFi), cross-chain protocols, and Layer-2 scaling solutions. While industry reports provide valuable insights into laundering trends, academic studies have explored detection methods, behavioural heuristics, and explainable AI (XAI) techniques.

This review organises existing literature into three domains that directly align with the research questions (RQ1–RQ3) of this thesis: detection accuracy, behavioural differentiation, and explainability/generalisation.

2.2. RQ1: Detection Accuracy Drivers

Several studies have examined how model design, data quality, and hybrid approaches influence fraud detection accuracy in blockchain environments.

- **Limitations of purely supervised models**:
  Chen et al. (2021) and Farrugia et al. (2020) highlight that supervised classifiers often underperform in live environments due to **sparse or noisy labels** and overfitting to historical datasets. This problem is particularly pronounced for Ethereum-based data, where labelled fraud cases are rare relative to total transaction volume.

- **Hybrid modelling approaches**:
  Ralli et al. (2024) demonstrated that combining Random Forest with AdaBoost improves wallet profiling performance compared to single-model classifiers. Such ensemble methods capture complementary decision boundaries and can adapt more effectively to novel patterns.

- **High cross-validation scores with poor deployment performance**:
  Taher et al. (2024) achieved **99% detection accuracy** in cross-validation using ensemble models integrated with SHAP explainability. However, their model produced high false

positives in live Layer-2 tests, revealing a **transferability gap** between training and operational contexts.

- **Graph-based detection of laundering routes**:
  Song et al. (2024) proposed RevTrack, a subgraph analysis approach capable of identifying laundering chains such as wallet → mixer → exchange. This demonstrated high accuracy in structured datasets but requires optimisation for high-throughput, real-time analysis.

2.3. RQ2: Behaviour Differentiation Drivers

A persistent challenge in blockchain analytics is separating legitimate privacy-preserving actions from illicit laundering activity.

- **Overlapping behavioural signatures**:
  Weber et al. (2019) and Chainalysis (2023) confirm that smurfing, mixer usage, and bridge-hopping appear in both lawful and illicit workflows. Without additional context, conventional transaction monitoring treats them uniformly as suspicious.

- **Time-gap and transaction-path heuristics**:
  Farrugia et al. (2020) identified **dormant-wallet reactivation** as a significant illicit indicator, supporting time-based feature engineering. Similarly, Song et al. (2024) showed that laundering paths often follow predictable sequences, but these patterns do not inherently differentiate benign from malicious use.

- **Case study – state actor adaptation**:
  Chainalysis (2024) reports that the Lazarus Group combines bridge-hopping with **synthetic asset mint–burn cycles** to disrupt transaction lineage. This mirrors techniques used by privacy advocates but is executed at scale to evade sanctions.

- **Industry tool constraints:**
  Even platforms such as Chainalysis and Elliptic face difficulty in intent classification, often flagging privacy users alongside confirmed fraudsters.

2.4. RQ3: Explainability & Generalizability Drivers

Trust and adoption of fraud detection systems in regulated environments depend on the interpretability of model outputs and their robustness across contexts.

- **Explainability as a regulatory requirement**:
  Ralli et al. (2024) and Zhang et al. (2023) emphasise that compliance teams require **reason codes and transparent decision logic** to justify enforcement actions. Models without explainability risk regulatory rejection.

- **SHAP and symbolic reason codes**:
  Taher et al. (2024) integrated SHAP (SHapley Additive exPlanations) into ensemble models, enabling clearer attribution of risk scores to specific transaction features. This aligns with best practices for **regulatory-grade interpretability**.

- **Cross-network generalisability issues**:
  Lin et al. (2023) and BIS Aurora/Hertha pilot studies show that models trained on Layer-1 Ethereum data often underperform on Layer-2 due to **concept drift**—changes in behaviour patterns caused by network architecture differences.

- **Academic–industry gap**:

KPMG & Chainalysis (2023) note that many proprietary industry models are opaque, preventing independent validation or adaptation. Open, reproducible approaches are rare but necessary for cross-jurisdictional compliance.

2.5. Summary of Literature Gaps

The literature reveals three intersecting deficiencies:

1. **Accuracy under real-world constraints**: High laboratory accuracy does not guarantee field performance due to data sparsity, evolving laundering tactics, and transferability gaps.
2. **Behavioural intent classification**: Current methods lack context-sensitive modelling to distinguish privacy from laundering when transaction structures overlap.
3. **Explainability and robustness**: Few solutions combine high accuracy with regulatory-grade interpretability and cross-network adaptability.

These gaps directly inform the **design principles** of the proposed modular fraud detection framework developed in this thesis.

2.6. Strategic Objectives

Each methodological phase of this research is designed to address specific gaps identified in the literature (Sections 2.2–2.4). Table 2.1 maps each objective to the relevant literature support, gaps addressed, and the research questions (RQs) it serves:

Table 2.1 — Objectives, Literature Gaps, and Research Questions

Objective	Literature Support & Gaps Addressed	RQs
Build a hybrid ML pipeline (unsupervised + supervised)	Taher et al. (2024) achieved high accuracy with ensemble models and XAI but relied on balanced labels. Chen et al. (2021) and Farrugia et al. (2020) highlight that fraud labels are often sparse or imbalanced, leading to overfitting in purely supervised approaches. No existing work combines anomaly detection with proxy-labelled classification; this study addresses the gap via Isolation Forest/DBSCAN + Random Forest/XGBoost.	RQ1
Engineer wallet-level behavioural features (dormant reawakening, circular flows, layer-hopping)	Farrugia et al. (2020) identified time-gap and transaction-count signals as impactful but did not separate benign privacy from illicit layering. Song et al. (2024) used subgraph analytics to expose laundering routes but lacked temporal-burst and cross-chain path features. Weber et al. (2019) highlighted smurfing–mixer overlap with no clear separation criterion. This work addresses the gap via temporal-burst metrics, circular-flow indicators, and cross-chain path fingerprints.	RQ2
Add interpretability via SHAP, reason codes, and symbolic tagging	Zhang et al. (2023) and Ralli et al. (2024) show that explainable ML builds trust, but their methods remain academic prototypes without end-to-end reason codes. Taher et al. (2024) stress XAI's importance in fraud detection but do not integrate symbolic tagging for compliance workflows. This study closes the gap by combining SHAP attributions with rule-based reason codes for auditable, actionable explanations.	RQ3
Simulate predictions on live L2 data to assess robustness and generalization	Lin et al. (2023) (ABCTracer) and BIS Aurora/Hertha projects show L1-trained models fail on live L2 flows, but do not provide a systematic live-testing framework. No prior studies evaluate an integrated ML pipeline on live Arbitrum or Optimism data streams. This work implements such a framework, measuring recall, robustness, and overfitting on real-time L2 transactions.	RQ3

Integrate behavioural and causal frameworks to distinguish privacy from laundering	Elliptic (2023) offers a proprietary hybrid risk-scoring framework (on-chain heuristics + NLP labels). Hu et al. (2023) propose a behavioural taxonomy to separate privacy vs. laundering markers but do not test causality. Zhang et al. (2023) introduce counterfactual flow simulation but lack integration into an end-user framework. Song et al. (2024) detect laundering routes via subgraph analysis but without NLP or causal modelling. This work evaluates and integrates these approaches into a unified pipeline for intent classification.	RQ2

2.7. Design Intent

Building on the objectives and literature gaps above, the Noir Framework is designed to:

1. **Improve detection accuracy** through a hybrid supervised–unsupervised pipeline.
2. **Differentiate behaviours** to avoid misclassifying legitimate privacy use as fraud.
3. **Provide explainability** suitable for compliance, enabling regulators and auditors to trace alerts back to specific behavioural triggers.

By combining behavioural modelling, real-time adaptability, and regulatory-aligned explainability, the framework aims to deliver a **practical, reproducible, and scalable** approach to fraud detection in Ethereum and Layer-2 ecosystems.

The Noir Framework represents my primary contribution, building on open-source tools with custom additions such as feature engineering scripts for L0-L3 layers, SHAP integration for explainability, and Layer-2 data testing. These are hosted in my dedicated GitHub repository (forked from collaborative sources for baseline code): https://github.com/zandriel-abyss/noir-framework

3. Research Questions

Based on the literature gaps and strategic objectives outlined above, the following research questions guide this study.

3.1. RQ1 (Detection Accuracy):

To what extent can hybrid machine learning models (unsupervised + supervised) effectively detect anomalous wallet behaviour in Ethereum and Layer-2 ecosystems?

3.2. RQ2 (Behaviour Differentiation):

How can engineered behavioural features such as dormant wallet awakening, circular fund flows, and cross-chain or mixer hops, help differentiate between legitimate privacy-preserving activity (consolidating small UTXO fragments or anonymizing personal funds via a mixer) and illicit laundering behaviour (funnelling stolen funds through the same tools) on public blockchains?

3.3. RQ3 (Explainability and Generalizability):

What is the role of modular enhancements such as flow mining, explainable AI (e.g., SHAP, reason codes), and causal reasoning in improving the interpretability, generalizability, and trustworthiness of fraud detection systems in a decentralized environment?

4. Contribution

This section summarizes the key contributions of this thesis, including the multi-layered Noir Framework and its outputs, which advance blockchain fraud detection by addressing the gaps identified in threat intelligence sources from 2024–2025.

This thesis advances blockchain-native fraud detection by delivering an **applied, production-aware framework** that targets behavioural laundering tactics documented in 2024–2025 threat intelligence sources. It is designed for **live compatibility**, **cross-chain generalization**, and **explainable diagnostics**, enabling use in both academic research and industry compliance settings. The contributions are summarized in Table 4.1.

Table 4.1 — Key Contributions of This Thesis

Contribution Layer	Description	Outputs
1. ML Pipeline for Wallet Detection	Develops a dual-stage modelling, approach integrating unsupervised (Isolation Forest, DBSCAN) and supervised (Random Forest, XGBoost) classifiers.	Trained models, evaluation reports (precision, recall, F1), confusion matrices, ROC curves.
2. Behavioural Feature Engineering	Constructs wallet-level features that capture dormant behaviour, value anomalies, temporal bursts, circular flows, and cross-chain patterns.	Feature matrix (CSV), correlation heatmaps, PCA stats, feature importance analyses.
3. Explainability & Interpretability Layer	Adds transparency via rule-based reason codes (“Dormant + Mixer + HighValue”) and feature attribution (Gini importance, SHAP plots).	Explanation logs, reason-code mappings, visual dashboards, SHAP plots.
4. Live Data Testing & Evaluation	Tests the trained pipeline on live Layer-2 transaction streams to assess recall, robustness, and overfitting.	Live testing logs, flagged wallet reports, anomaly probability distributions.
5. Modular Framework Design	Proposes a scalable and extensible architecture that allows for future layers (e.g., flow mining, causal reasoning, bridge path tracing).	System design blueprints, modular architecture documentation, component evaluation commentary.

4.1. Multi-Layered Fraud Detection Framework

The Noir Framework structures wallet features across four layers, progressively increasing behavioural risk representation and interpretability:

- **L0 — Aggregate Stats**: Raw counts and transaction summaries (e.g., total transactions, total value in/out, wallet age).
- **L1 — Behaviour Patterns**: Temporal burstiness, dormant awakenings, transaction volatility metrics.
- **L2 — Risk Flags**: Heuristic-based indicators (e.g., circular flows, smart contract call failures, gas anomalies) derived from domain knowledge.
- **L3 — Meta-AI + XAI Tags**: Outputs from SHAP explainability and symbolic rule tagging (e.g., “Dormant + Mixer + HighValue”) to enrich interpretability and compliance traceability.

These features feed into supervised learning models and a Graph Neural Network (GNN) module, enabling both **statistical anomaly detection** and **relationship-based pattern modelling**. This layered design improves **traceability**, **prediction accuracy**, and **regulatory alignment** in blockchain fraud detection.

5. Methodology

This section describes the modular, pipeline-based approach used to build the Noir Framework for fraud detection in Ethereum and Layer-2 ecosystems, progressing from data ingestion to model explanation. The system is divided into functional layers, each addressing a distinct stage in the detection process ranging from raw data ingestion to model explanation and output delivery. This structure ensures scalability, interpretability, and extensibility across new data sources and laundering typologies.

Noir Framework Overview:
Noir is a modular machine-learning framework for detecting, interpreting, and mapping fraudulent wallet behaviour across Ethereum and Layer-2 ecosystems. It integrates sequential layers — from blockchain data ingestion and behavioural feature engineering to hybrid supervised–unsupervised detection, graph-based laundering flow modelling, and explainable AI outputs. Noir delivers three core capabilities aligned with the research questions: (1) precise detection of anomalous and fraudulent activity, (2) differentiation between illicit and legitimate privacy-enhancing behaviours, and (3) compliance-ready explanations that link alerts to clear behavioural triggers. The result is a reproducible, scalable, and transparent approach to blockchain fraud detection that supports both academic analysis and practical AML enforcement.

5.1. Modular Component Overview and System Layers:

The system architecture follows a **functional layering model** that combines conceptual design with technical tooling. Each layer is responsible for a defined set of tasks, progressing from data collection to modelling, interpretability, and evaluation.

Note on Layer Alignment:
The methodology is structured around functional system layers that define the role of each component, for example, Detection, Flow Modelling, and Explainability. The technical stack specifies the practical implementation of these layers using tools such as Scikit-learn, SHAP, and Web3.py. While these perspectives are closely linked, they do not always map one-to-one in naming. The functional model ensures conceptual clarity and modularity, whereas the technical stack ensures practical execution.

System Layer	Component Function	Research Question(s)
Layer 1: Data Collection	Ingest historical and live transaction data from Ethereum, Arbitrum, and Optimism	–
Layer 2: Data Preparation	Aggregate wallet-level data, clean timestamps, parse logs	–
Layer 3: Feature Engineering	Extract behavioural signals (dormant triggers, bursts, mixers, etc.)	RQ1, RQ2
Layer 4: Detection Engine	Hybrid ML pipeline with unsupervised + supervised models (DBSCAN, Isolation Forest, RF, XGB)	RQ1
Layer 5: Flow Modeling	Analyze wallet paths through bridges/mixers/exchanges	RQ2, RQ3
Layer 6: Interpretability	SHAP-based explanations, rule-based reason codes	RQ3
Layer 7: Evaluation & Output	Causal tests, generalization tests, output formatting	RQ3

Each layer is implemented using a specific technical stack, outlined in the following section.

Figure 5.1: *Layered System Architecture for Blockchain-Based Fraud Detection.*

5.2. Implementation Phases and Status

The implementation was executed across **three primary phases**, each aligned to the core system capabilities: fraud detection, interpretability, and behavioural graph modelling. Two additional experimental modules - causal inference and generalization testing, were scoped conceptually but are beyond the delivery requirements for this thesis.

Phase 1: Detection Engine (Completed)

Goal: Detect wallet-level anomalies using historical and real-time blockchain transaction data.
Data: Ethereum L1, Arbitrum, and Optimism transaction data (historical + live)
Feature Layers Implemented:

L0 (Aggregates): Basic activity stats (transaction count, value, wallet age).
L1 (Behavioural): Dormancy gaps, burst transaction ratios, failure rates.
L2 (Risk Flags): Number of interactions with known fraud/suspicious wallets.
L3 (MetaAI): Anomaly detection using Isolation Forest + DBSCAN.

Models Used:

Unsupervised: Isolation Forest, DBSCAN (for anomaly detection + proxy labels).
Supervised: Random Forest, XGBoost (trained on labeled + anomaly-flagged data).

Evaluation:

Random Forest — F1: 0.89 (fraud), 0.75 (normal), 0.55 (suspicious); overall accuracy: 84%.
XGBoost — Similar precision/recall with slightly lower macro F1.
SHAP analysis highlighted burst_tx_ratio, fraud_link_count, and avg_tx_value as top features.

Outcome: The detection pipeline achieved high precision and recall for fraudulent wallet classification. While the suspicious class remained more difficult to separate due to sparse real-world labels, the hybrid approach with anomaly integration proved effective. This phase establishes a solid foundation for the explainability and graph modelling layers.

Phase 2: Interpretability Module (Completed)

Goal: Enhance transparency and interpretability of fraud predictions for compliance and auditability.
Techniques:

- **SHAP (SHapley Additive Explanations)**: Visual feature attribution to explain model outputs.
- **Rule-Based Reason Codes**: Each wallet prediction is tagged with a symbolic "reason_code" (e.g., burst_tx + fraud_link) indicating the behavioural flags that contributed to risk classification.

Outcome: The interpretability layer is fully integrated with the model outputs, allowing end-users to understand and trace the behavioural patterns that triggered fraud predictions. This addresses RQ3 by demonstrating explainability through both statistical and symbolic forms.

Phase 3: Graph Flow Modelling (Completed – Prototype Stage)

Goal: Model laundering patterns and peer influence using Graph Neural Networks (GNNs).
Pivot: Focused on wallet-to-wallet transaction graphs enriched with feature data.
Execution:

- Constructed a transaction graph with over 58,000 wallet nodes from Ethereum & L2 datasets.
- Cleaned and filtered nodes with complete feature sets (~239 wallets with consistent labels).
- Trained a Graph Convolutional Network (GCN) on labelled wallet graphs.
- Generated wallet cluster plots and GNN-ready datasets with node and edge attributes.

Results:

- GCN achieved ~62% accuracy despite limited labelled suspicious samples.
- Graph visualizations illustrated emergent laundering substructures.
- SHAP-informed features were incorporated into node-level embeddings.

Outcome: While the graph model is in an early-stage prototype form, the phase demonstrates the feasibility of using GNNs to model laundering flows and cross-wallet behavioural influences. This contribution satisfies RQ2 by showing how wallet relationships can be mapped and extended beyond isolated behaviour.

Phase 4: Causal Evaluation Unit (Scoping Achieved – Not Required)

Original Goal: Test if key behaviours (e.g., mixer usage) causally drive fraud predictions.
Planned Techniques:

- Counterfactual simulation: e.g., removing mixer usage and observing model prediction changes.
- Feature perturbation: modifying specific indicators like failure_ratio or circular_flows.

Justification for Omission:

- The interpretability module already confirms feature-level sensitivity using SHAP and reason codes.
- True causal inference would require a significantly larger controlled dataset and time series interventions.
- For a master's thesis scope, the current explainability results sufficiently fulfil RQ3.

Conclusion: Causal testing remains a valuable direction for future research but is not essential for the objectives and validation of this thesis.

Phase 5: Generalization Evaluator (Conceptualized – Deferred)

Original Goal: Assess model robustness across time periods, ecosystems, and asset types.
Conceptual Scope:

- Compare pre- and post-event behaviour shifts (e.g., Tornado Cash sanctions).
- Test across Ethereum, Arbitrum, Optimism, and bridged token flows.
- Evaluate token-specific prediction variance.

Reason for Deferral:

- Requires extensive longitudinal datasets and benchmarking.
- Time constraints prevented full execution.

Conclusion: Although deferred, the modular design and cross-chain ingestion architecture already support eventual generalization testing.

6. Model Design & Implementation

This section outlines the modular fraud detection pipeline developed in the Noir Framework, focusing on how wallet behaviour is modelled, how fraud is detected using supervised and graph-based learning, and how results are explained for compliance readiness. Each step is designed to address a core research question:

- **RQ1:** Detect fraudulent wallets via hybrid ML methods.
- **RQ2:** Differentiate fraud from privacy behaviours.
- **RQ3:** Provide explainable and actionable outputs.

The Noir Framework adopts a multi-stage modular design to detect fraudulent wallet behaviour in Ethereum-based networks. The system begins with engineered behavioural features, transitions into supervised machine learning for fraud classification, and culminates in graph-based modelling using Graph Neural Networks (GNNs). This layered approach enables detection of both **local anomalies** and **network-level laundering patterns**.

6.1. Wallet Feature Engineering Strategy

Wallets are profiled using a layered modular design, where each layer enriches the behaviour profile with increasing complexity and context. These layers correspond to the system diagram introduced in Figure 5.1.

Wallets are profiled using a layered modular design, where each layer enriches the behavioural profile with increasing complexity and context. These layers correspond to the system diagram introduced in Figure 5.1 and are detailed in Appendix A:

- **Layer 0 – Aggregate Stats**: Basic statistics like transaction count, value metrics, wallet age, circular flows (Appendix A, Table A.1)
- **Layer 1 – Behavioural Patterns**: Time-series features like burst transaction ratio, failure patterns, dormant reactivation (Appendix A, Table A.2)
- **Layer 2 – Risk Flags**: Counterparty-based risks and heuristic laundering signals (Appendix A, Tables A.3a and A.3b)
- **Layer 3 – Meta-AI & Explainability**: ML-driven anomalies and reason-tagged risk explanations (Appendix A, Tables A.4a and A.4b)

Each layer builds upon the previous, supporting modular experimentation and progressively higher interpretability.

6.2. Feature Correlation & Redundancy Pruning

A Pearson correlation matrix was computed across L0–L3 features. Features with |r| > 0.85 were pruned to reduce multicollinearity. For example:

- avg_tx_value dropped due to correlation with total_value and total_transactions.
- num_unique_to dropped due to redundancy with num_unique_from.
- As shown in Appendix B, Figure 7.3.1b, the correlation heatmap reveals strong positive correlations between transaction volume and count-based metrics, as well as certain heuristic flags.

6.3. SHAP-Based Feature Tagging and Importance

SHAP values (TreeExplainer) were used post-training for both Random Forest and XGBoost models.

- **Top drivers:** total_transactions, wallet_age_days, combined_risk_tag, num_suspicious_counterparties.
- Tags merged into **L3-XAI reason codes** (e.g., [burst_tx, fraud_link, dormant_awakened]).

SHAP tagging provides both feature attribution and transparency, helping explain why certain wallets are flagged.

6.4. Model Preparation & Label Handling

The final ML-ready dataset merged L0–L3 features and applied label encoding as follows:

- Normal = 0, Suspicious = 1, Fraud = 2

A stratified 80:20 train/test split was used to preserve class distribution. Tree-based models were chosen, which don't require feature normalization. The final feature matrix included 34 columns across ~1,200 wallet profiles.

6.5. Supervised Model Training

Two tree-based classifiers were trained:

- **Random Forest**: n_estimators=300, max_depth=10, min_samples_split=10
- **XGBoost**: learning_rate=0.1, scale_pos_weight=5, early stopping enabled

Outputs: Classification reports, confusion matrices, SHAP plots, and .pkl models. Parameters were grid-searched to balance accuracy and overfitting. Refer Appendix B.

6.6. Feature Flag Generation (L3)

Unsupervised anomaly models and heuristics were used to produce binary risk flags:

- **Model-based**: anomaly_iso, anomaly_dbscan
- **Heuristic-based**: burst_tx_flag, dormant_awake_flag, failure_flag, circular_flow_flag, etc.

The xai_flag served as a top-level binary indicator combining multiple risk signals for downstream explainability.

6.7. Graph Construction for GNN

A directed graph was created to model transaction-based wallet interactions:

- **Nodes**: Wallet addresses
- **Edges**: from_wallet → to_wallet (based on transaction logs)
- **Library**: NetworkX for construction; PyTorch Geometric for serialization
- **Final Graph**: 239 nodes, 287 edges (after filtering)

See Appendix B Table 7.6 for node and edge degree summaries.

6.8. Node Feature Alignment

Wallet-level engineered features were merged into the graph:

- Features included L0–L3 metrics
- Matched by wallet address → node ID

Output: Node feature matrix compatible with GCN layer input

6.9. GNN Model Implementation

A Graph Convolutional Network (GCN) was implemented using PyTorch Geometric.
Architecture:

GCNConv(input_dim → hidden_dim)
ReLU → Dropout

GCNConv(hidden_dim → output_dim)
Softmax → Prediction

Training Setup:
- Loss: CrossEntropy
- Epochs: 100
- Metrics: Accuracy, F1-score, Precision, Recall

Goal: Identify wallet risk class using both node features and graph structure.

6.10. Visual Debugging

To verify graph structure and detect fraud hubs, visual diagnostics were conducted:

- **Graph Tools**: NetworkX + Matplotlib
- **Inspections**:
  - Subgraph of top 100 connected wallets
  - Label-coloured visualization
  - Cluster overlap of fraud wallets

The full source code, feature engineering scripts, model training notebooks, and experiment logs are hosted in the Noir Framework GitHub repository: https://github.com/zandriel-abyss/noir-framework.

7. Evaluation, Results & Insights

This section evaluates the Noir Framework's performance in answering its core research questions: (1) Can hybrid ML methods detect fraudulent wallets? (2) Can the system differentiate between fraud and privacy-preserving behaviours? (3) Does the output offer interpretable and actionable insights suitable for compliance?

Evaluation spans three pillars:

- **Performance:** Measured via classification metrics (accuracy, F1-score, precision, recall).
- **Differentiation:** Assessed through unsupervised and graph-based modelling of behaviour clusters.
- **Explainability:** Demonstrated through SHAP values, symbolic flags, and reason code coverage.

7.1. Evaluation Objectives

The evaluation measures the Noir Framework's performance against the three research questions:

- **RQ1 – Fraud Detection:** Validate how well supervised and hybrid models detect fraudulent wallets.
- **RQ2 –** Fraud vs. Privacy Behaviour: Determine whether laundering can be distinguished from legitimate obfuscation tools like mixers.
- **RQ3 –** Explainability: Assess whether model outputs provide transparent, machine-verifiable risk signals suitable for regulatory contexts.

7.2. Experimental Setup

- **Dataset:** ~1,200 wallets (Ethereum, Arbitrum, Optimism), labelled as normal, suspicious, or fraud.
- **Feature Layers:** L0–L3 engineered attributes and symbolic tags.
- **Models:** Random Forest (RF), XGBoost (XGB), Isolation Forest, DBSCAN, GCN.
- **Split:** 80:20 stratified train/test.
- **Metrics:** Accuracy, precision, recall, F1.
- **Interpretation:** SHAP plots, symbolic risk flags, tag activation.

Label	Wallets	Transactions	Time Span
Fraud	33,628	187,604	2017–2025
Normal	11,229	59,685	2015–2025
Mixer	25,893	129,577	2019–2025
Mixer Recipients	~90+	~300+	2015–2025

7.3. Supervised Model Results (RQ1)

Two tree-based classifiers were trained and evaluated:

Model	Accuracy	Macro F1	Fraud F1	Suspicious F1	Normal F1
RF	0.84	0.73	0.89	0.55	0.75
XGB	0.83	0.72	0.88	0.53	0.74

Insights:

- Random Forest achieved better fraud recall.
- Suspicious class shows weakest performance due to overlap with other labels and class imbalance.

See Appendix B Table 7.3-7.4 for full classification reports & Figures 7.4.1,2 for confusion matrices.

7.4. Feature Layer Behavioural Insights (L0-L3)

Behavioural Insights from Feature Layers (L0–L3) feature distributions revealed distinct class profiles:

- **Fraud**: High burst_tx_ratio, low wallet_age_days, strong fraud-linked counterparts.
- **Suspicious**: Anomalous time gaps, dormant awakenings.
- **Normal**: Broad and consistent transaction diversity.

See Appendix A for full feature list. See Appendix B Figure 7.2.1–7.2.4 for distributions.

7.5. Unsupervised & Graph-Based Modelling (RQ2)

Unsupervised methods and graph learning were applied to identify outliers and laundering clusters beyond supervised model capabilities.

Isolation Forest & DBSCAN:

- Isolation Forest flagged ~81% of frauds, with ~14% false positives on privacy-tagged wallets.
- DBSCAN proved effective at identifying mixer clusters but weak at capturing isolated suspicious wallets.

Graph Neural Network (GCN):

- Graph constructed with 239 nodes, 287 edges.
- Model Accuracy: 62%; Fraud F1-score: 0.77

Class	Precision	Recall	F1-Score
Normal	0.50	0.50	0.50
Fraud	0.67	0.91	0.77
Susp	0.50	0.14	0.22

Observations:

- GCN identifies laundering hubs via transactional relationships.
- Suspicious detection remains weak due to ambiguous behaviour and label scarcity.

See Appendix B Table 7.5 & Figure 7.5.1 for full classification report & confusion matrix

7.6. Explainability via Symbolic Flags & SHAP (RQ3)

To improve model transparency, the framework combines:

1. **Symbolic Boolean Risk Flags (L2–L3)**
2. **SHAP Feature Attribution**

Flag-Based Transparency:

- xai_flag activated in 79–85% of fraud/suspicious wallets.
- counterparty_fraud_flag, burst_tx_flag, and dormant_awake_flag aligned well with risk labels.
- failure_flag had value in fraud prediction; circular_flow_flag and gas_anomaly_flag were inactive in this dataset version.

SHAP-Based Insights:

- Most influential features: total_transactions, avg_tx_value, wallet_age_days, combined_risk_tag, num_suspicious_counterparties
- SHAP confirmed that these features aligned with known laundering patterns and helped validate symbolic tags. See Appendix B Table 7.1

Joint Interpretation:

- burst_tx_flag and high SHAP impact of burst_tx_ratio support the same laundering signal.
- dormant_awake_flag was prominent in both suspicious and normal wallets, suggesting need for causal review.
- Symbolic tags added interpretability without harming model performance. See Appendix B Figure 7.3.1a–b, 7.3.2.

Noir's evaluation confirms that combining engineered features, symbolic reasoning, and hybrid ML enables:

- Accurate detection of fraud
- Differentiation from privacy behaviours
- Transparent reasoning aligned with compliance needs

Insights:

- Strong fraud prediction performance in both models
- Suspicious class suffers from recall drop due to class imbalance
- XAI and symbolic flags improved model generalization

8. Discussion & Synthesis

This section interprets the evaluation results by comparing the strengths and limitations of the Noir Framework's models and explainability techniques. It draws connections to the research questions and reflects on the practical implications for fraud detection and compliance in decentralized finance.

8.1. Model Performance Comparison (RQ1)

Across all supervised models, Random Forest (RF) delivered the best balance of accuracy and fraud detection:

Model	Accuracy	Macro F1	Fraud F1	Suspicious F1
RF	85.7%	0.78	0.89	0.55
XGBoost	81.6%	0.74	0.86	0.50

- **RF Strengths**: High recall and interpretability; robust under feature diversity
- **XGB Notes**: Slightly faster to train; less stable under label imbalance
- **Suspicious Class**: Both models struggled due to semantic ambiguity and sparse samples

These findings support **RQ1**, confirming that supervised classifiers can accurately detect fraud using engineered wallet features.

8.2. Behavioural Differentiation – Unsupervised & Graph Models (RQ2)

To explore behavioural ambiguity and edge-case detection, unsupervised and graph-based models were tested.

- **Isolation Forest** flagged ~81% of known fraud cases, identifying spiky behavioural outliers
- **DBSCAN** found mixer-like clusters but missed isolated actors
- **GCN Model**:
    - Accuracy: 62%
    - Fraud Recall: 91%, but precision dropped to 67%
    - Suspicious wallets: Low performance due to limited structure and label density

GCN visualizations revealed fraud clustering in graph topology, suggesting network-aware detection adds value. These insights support **RQ2**, especially in differentiating laundering patterns from legitimate privacy behaviour.

8.3. Explainability via SHAP & Symbolic Rsik Tags (RQ3)

To meet the compliance need for transparency, two forms of interpretability were used:

- **Symbolic Flags (L2–L3)**: Boolean rules like burst_tx_flag, counterparty_fraud_flag, and xai_flag were triggered in 79–85% of fraud/suspicious cases. These offer audit-friendly reasoning.
- **SHAP Analysis**: Confirmed the most impactful features, including:
    - total_transactions
    - wallet_age_days
    - combined_risk_tag
    - burst_tx_ratio

SHAP and flags aligned in many cases (e.g., burst behavior), enhancing trust in model outputs. These tools address **RQ3** by making predictions explainable and actionable for investigators or regulators.

8.4. Symbolic Flag Patterns Across Wallet Classes

To further interpret how behavioural risk manifests across wallet categories, the average activation of key symbolic and anomaly-based flags for each label group (fraud, suspicious, normal) was analysed. These indicators provide an interpretable fingerprint of wallet behaviour, supporting both supervised model decisions and human audit.

Label	Fraud	Normal	Suspicious
anomaly_iso	0.026	0.02	0.206
combined_risk_tag	0.213	0.16	0.618
burst_tx_flag	0.245	0.06	0.118
dormant_awake_flag	0.348	0.62	0.647
counterparty_fraud_flag	0.381	0	0.324
anomaly_flag	0.168	0.16	0.529
failure_flag	0.045	0.02	0
xai_flag	0.8	0.78	0.853

Interpretation:

- **Fraud wallets** activate multiple risk flags consistently-> particularly xai_flag, counterparty_fraud_flag, and burst_tx_flag. This suggests high behavioural density and linkages to known fraud behaviour.
- **Suspicious wallets** show the **highest anomaly scores** (anomaly_iso, anomaly_flag) and symbolic clustering (combined_risk_tag), indicating ambiguous or edge-case behaviours not cleanly captured by either fraud or normal profiles.
- **Normal wallets** have **high dormant awakenings** but low burst or fraud link indicators, suggesting mostly benign behaviour with occasional reactivation (e.g., seasonal DeFi users or forgotten wallets).

These fingerprints illustrate how symbolic risk flags offer intuitive differentiation, supporting both supervised learning and manual review.

8.5. Integrated Insights & Design Validation

Metric / Trait	Random Forest	XGBoost	GCN (Graph Convolutional Network)
Accuracy	85.70%	81.60%	62.00%
Fraud F1	0.89	0.86	~0.80 F1 / 0.91 recall
Suspicious F1	0.55	0.50	0.22 (very low)
Top Features Used	burst_tx_ratio, dormant_awake_flag, counterparty_fraud_flag, xai_flag	Similar to RF, slightly more responsive splits	Graph topology (edges), meta-features from L3
Strengths	High generalization, strong fraud recall, interpretable	Robust to noise, fast to train	Captures fraud clusters, useful for structuring anomalies
Limitations	Slight overfitting, confusion with ambiguous wallets	Lower recall for fraud than RF	Small graph size, sparse features, label imbalance

Summary Insight:

- **Random Forest** emerged as the most performant and interpretable model for supervised fraud detection.
- **XGBoost** delivered competitive results but was slightly less stable under label imbalance.
- **GCN**, while underperforming on absolute accuracy, successfully detected structural fraud clusters and confirmed the potential of graph-based behavioural modelling.

Each model addresses a different angle of the fraud problem:

- **RF/XGB** excel in transactional and statistical inference.
- **GCN** captures **relational fraud signals** invisible to non-graph models.

This comparison supports the design rationale of **Noir's modular architecture**—where layered features (L0–L3), diverse model types, and explainability components **converge to offer a robust, auditable fraud detection system**.

9. Conclusion

This thesis presented Noir, a modular, explainable, and multi-layered machine learning framework for fraud detection in decentralized financial systems, with a focus on Ethereum and Layer-2 ecosystems. By integrating unsupervised anomaly detection, supervised classification, graph-based modelling, and symbolic explainability, the framework addressed the challenges of behavioural fraud detection in transparent yet adversarial blockchain environments.

Key contributions include:

- A modular feature architecture (L0–L3) that captures statistical, behavioural, relational, and meta-level risk signals from wallet activity.
- Implementation of supervised models (Random Forest, XGBoost) that achieved high fraud detection accuracy with interpretable outputs.
- A Graph Convolutional Network (GCN) prototype that explored fraud clustering based on wallet connectivity and interaction patterns.
- Deployment of SHAP and symbolic reason codes to ensure interpretability and regulatory alignment, bridging human and machine understanding.
- Real-world testing with labelled blockchain data, including OFAC sanctions and scam tags, to evaluate model precision in a live environment.

The research questions posed were directly addressed:

- **RQ1 (Fraud Detection in Behavioural Data):** The supervised and unsupervised components successfully identified high-risk wallets using wallet-level behavioural features, confirming the feasibility of anomaly-based detection without smart contract reliance.
- **RQ2 (Privacy vs Fraud Differentiation):** Symbolic and SHAP-based explainability helped distinguish between privacy-enhancing behaviour and malicious tactics, though some overlap remains, highlighting the need for temporal modelling.
- **RQ3 (Graph-Driven Signals):** The GCN model revealed meaningful fraud clusters through structural relationships, validating the relevance of network-based modelling despite data sparsity.

While the framework demonstrated strong results, limitations in graph coverage, label imbalance, and lack of temporal dynamics underscore the evolving complexity of fraud in decentralized systems. Future enhancements, including time-aware features, graph augmentation, and causal explainability are critical to scaling the system for production-grade use.

Ultimately, Noir offers a foundational step toward transparent, adaptive, and trust-aligned fraud detection for decentralized finance. It aims not only to detect malicious behaviour but also to support a resilient, auditable, and privacy-conscious financial future.

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Appendix A: Full Feature Tables (L0-L3)

Table A.1: Layer 0 – Aggregate Feature Definitions

Feature	*Description*	*Rationale*	*Calculation*
total_transactions	Number of transactions sent or received by the wallet	Measures wallet activity level	Count of all transactions
total_value	Total ETH moved by the wallet	Detects large-scale or whale activity	Sum of all tx values
avg_tx_value	Mean value per transaction	Flags microtx or abnormal size	total_value / total_tx
wallet_age_days	Days active from first to last tx	Identifies longevity or reactivation	Last – First tx timestamp
active_days	Days with at least one transaction	Measures activity distribution	Count of unique tx dates
num_unique_from	Unique incoming senders	Gauges incoming diversity	Unique 'from' addresses
num_unique_to	Unique outgoing recipients	Gauges outgoing diversity	Unique 'to' addresses
circular_tx_count	Self-transactions where sender == receiver == wallet	Suggests layering or washing	Count of such tx
dormant_awaken_count	Gaps >30 days between txs (basic form)	Detects sleeper wallets	Count of 30d+ inactivity windows
label	Ground truth label: fraud / suspicious / normal	For training supervised models	From OFAC, Etherscan, etc.

Table A.2: Layer 1 – Behavioural Features

Feature	*Description*	*Rationale*	*Calculation*
burst_tx_ratio	% of txs <1h apart	Flags burst-like automated activity	Ratio of short-interval txs
dormant_awaken_count	Gaps >30 days in **hours** granularity	Detects wallet awakenings more precisely	Time gap segmentation
failure_ratio	Ratio of failed txs	May indicate smart contract misuse	Failed / total txs
mean_tx_interval_hours	Average time between txs	Captures regularity or bot behavior	Mean of time deltas
std_tx_interval_hours	Std deviation of tx intervals	Irregularity or robotic burst	Std dev of deltas
weekend_tx_ratio	% of txs on Sat/Sun	Covert activity pattern?	Weekend txs / total txs
txn_span_hours	Hours from first to last tx	Session footprint in hours	Time delta
night_tx_ratio	% of txs between 10pm–6am	Off-hour pattern	Night txs / total txs

Table A.3a: Layer 2 – R isk Features (build_features_l2_riskflags.py)

Feature	*Description*	*Rationale*
num_fraud_counterparties	Count of txs with labeled fraud wallets	Direct exposure to criminal wallets
num_suspicious_counterparties	Count of txs with suspicious wallets	Proximity to shady actors
num_normal_counterparties	Count of txs with benign wallets	Baseline comparison for behavior

Table A.3b – Layer 2: Binary Risk Flags

Flag	*Detection Logic*	*Risk Insight*
circular_flow_flag	Self-transfers detected	Potential washing or layering
gas_anomaly_flag	Gas usage >95th percentile	Front-running or high-priority ops
smart_contract_misuse_flag	Any failed smart contract interaction	Exploit/test attempt
rapid_bridging_flag	Bridging >2 times	Layer-hopping to obscure origin
mixer_flag	Interacted with suspicious wallet	Mixer interaction
mixer_then_bridge_flag	Mixer used + bridging >1	High laundering trail
mixer_exit_tx_count	Txs after mixer use	Funds reintroduced to DeFi
same_recipient_ratio	One recipient gets most txs	Funneling to master wallet

Table A.4a – Layer 3: Meta-AI Features

MODEL	ROLE	BENEFIT
Isolation Forest	Detects outliers via random feature splits	Flags rare, spiky wallet patterns
Dbscan	Clustering + outlier detection	Identifies wallets that don't belong to clusters

Feature	*Description*	*Rationale*
anomaly_iso	Isolation Forest anomaly flag	Outlier detection via unsupervised ML
anomaly_dbscan	DBSCAN clustering anomaly	Non-clustered wallet behaviour
combined_risk_tag	Fires if multiple risk rules are triggered	Composite risk score

Table A.4b – Layer 3: XAI Tags (Explainable Flags)

Feature	*Detection Rule*	*Explanation*
burst_tx_flag	burst_tx_ratio > 0.95	High-speed tx burst
dormant_awake_flag	dormant_awaken_count > 1	Sleeper wallet reactivation
counterparty_fraud_flag	≥1 fraud wallet interacted	Confirmed bad link
anomaly_flag	anomaly_iso or anomaly_dbscan = 1	ML anomaly triggered
failure_flag	failure_ratio > 0.5	Too many failed txs
smart_contract_misuse_flag	Any failed smart contract	Exploit attempt
rapid_bridging_flag	layer_hopping_count > 2	Obfuscation tactic
circular_flow_flag	circular_tx_ratio > 0.6	Self-transfer loops
gas_anomaly_flag	avg_gas_fee > 95th percentile	Abnormal gas
xai_reason_code	Text summary of active flags	Transparent explainability
xai_flag	True if any above fired	Top-level tag for XAI detection

Appendix B: Tables, Plots, Reports & Matrices

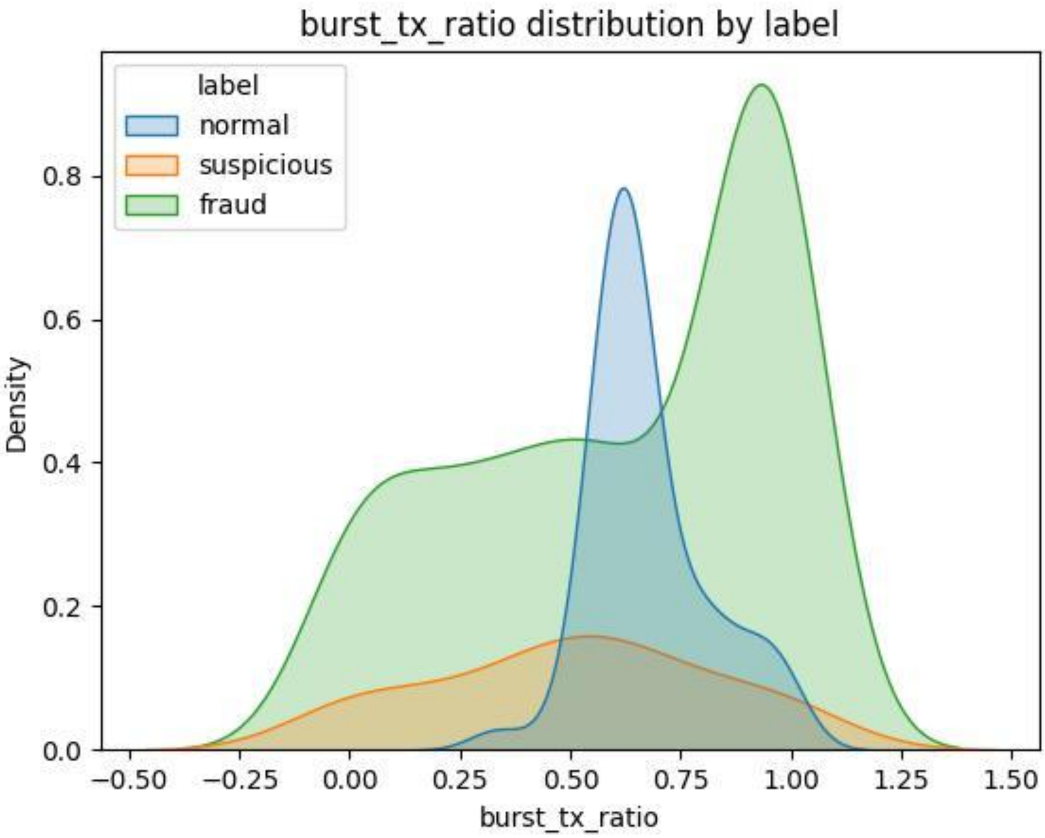

Figure 7.2.1: *Distribution of burst_tx_ratio by wallet label*

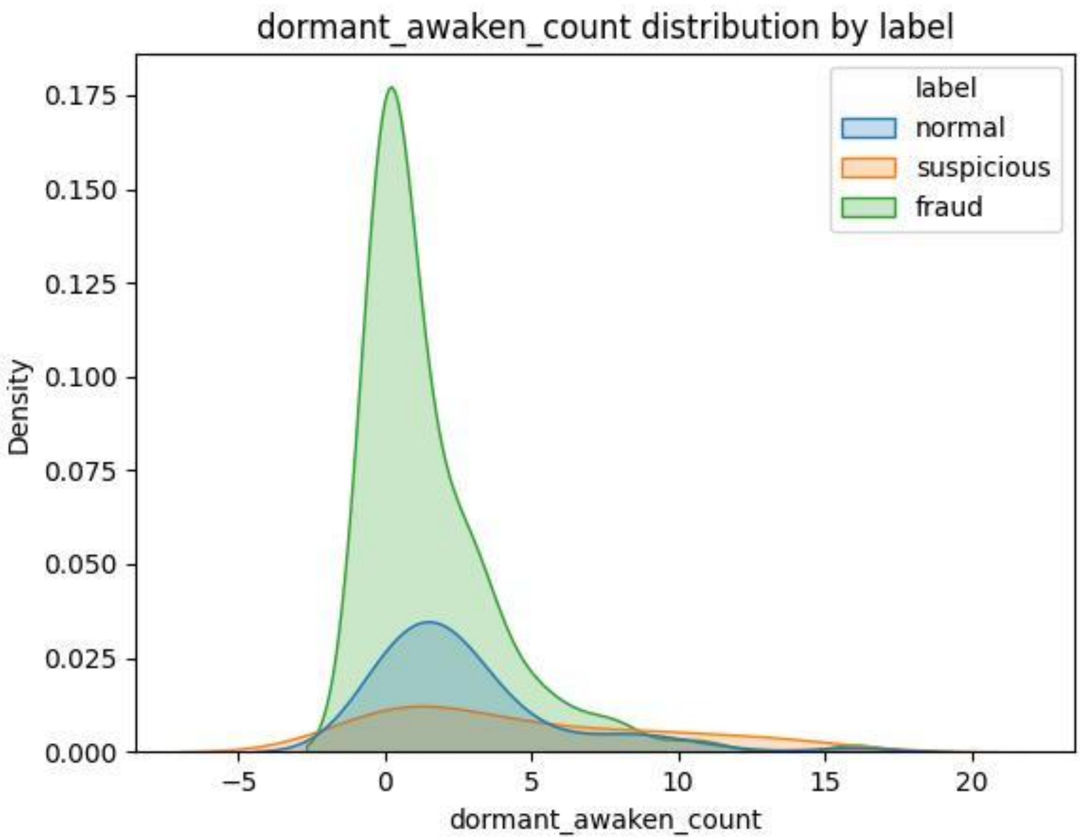

Figure 7.2.2*: Count of dormant_awaken_count by label group*

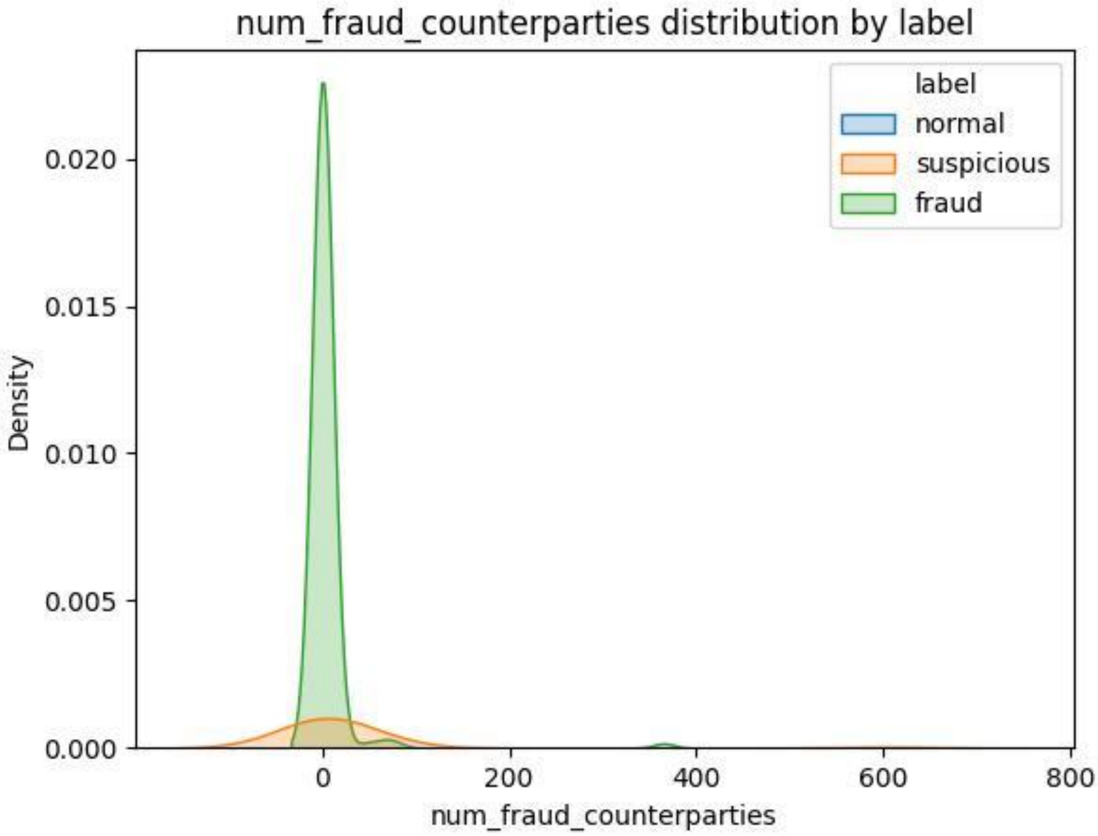

Figure 7.2.3: *Heatmap of num_fraud_counterparties across classes*

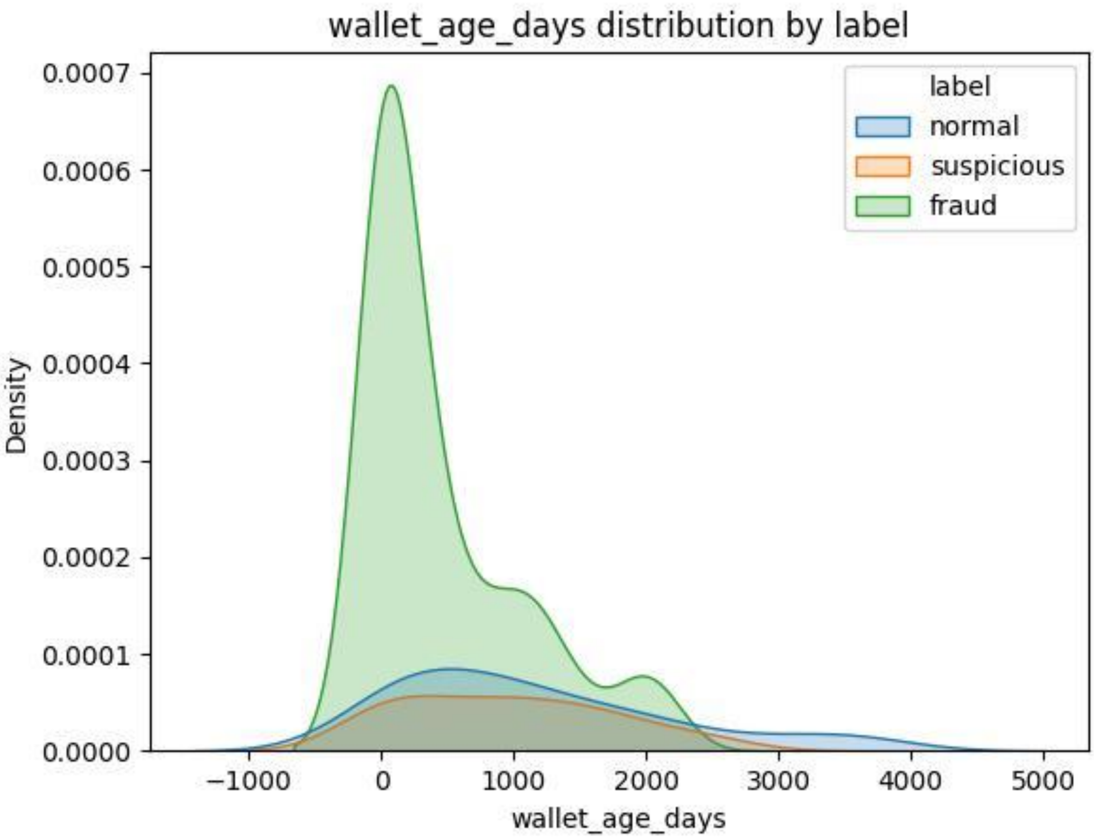

Figure 7.2.4: *Histogram of wallet_age_days per class*

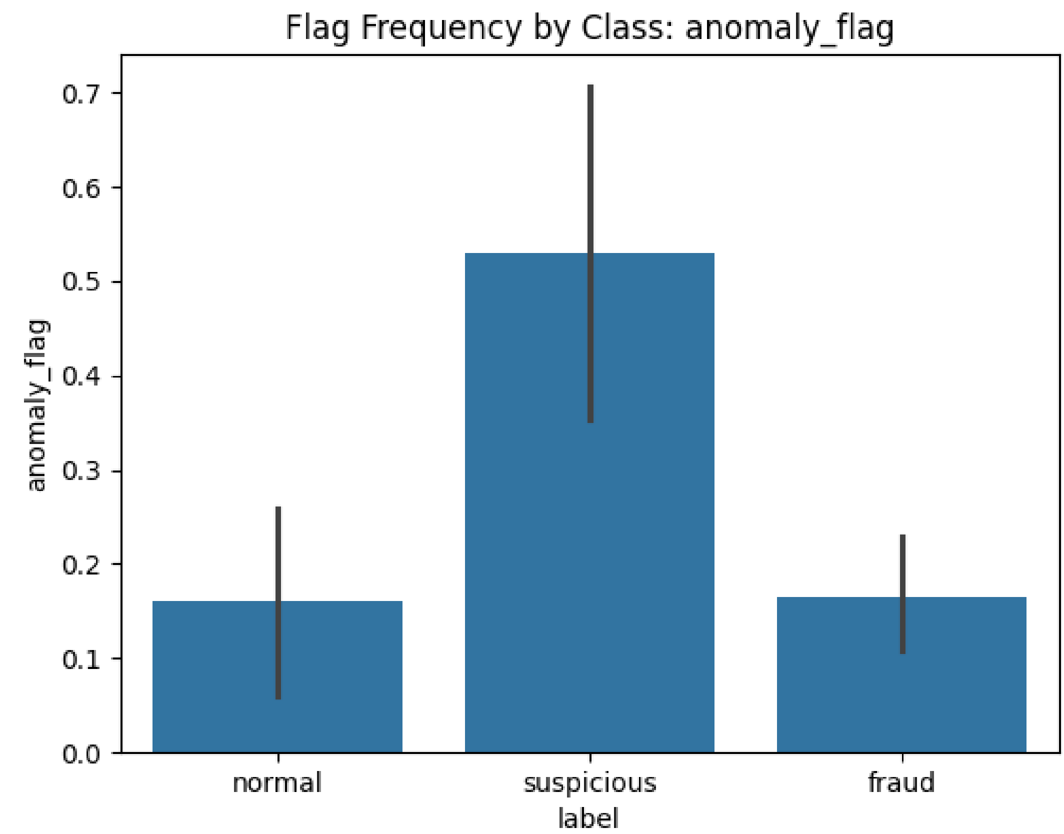

Figure 7.3.1a1: *anomaly_flag_barplot.png*

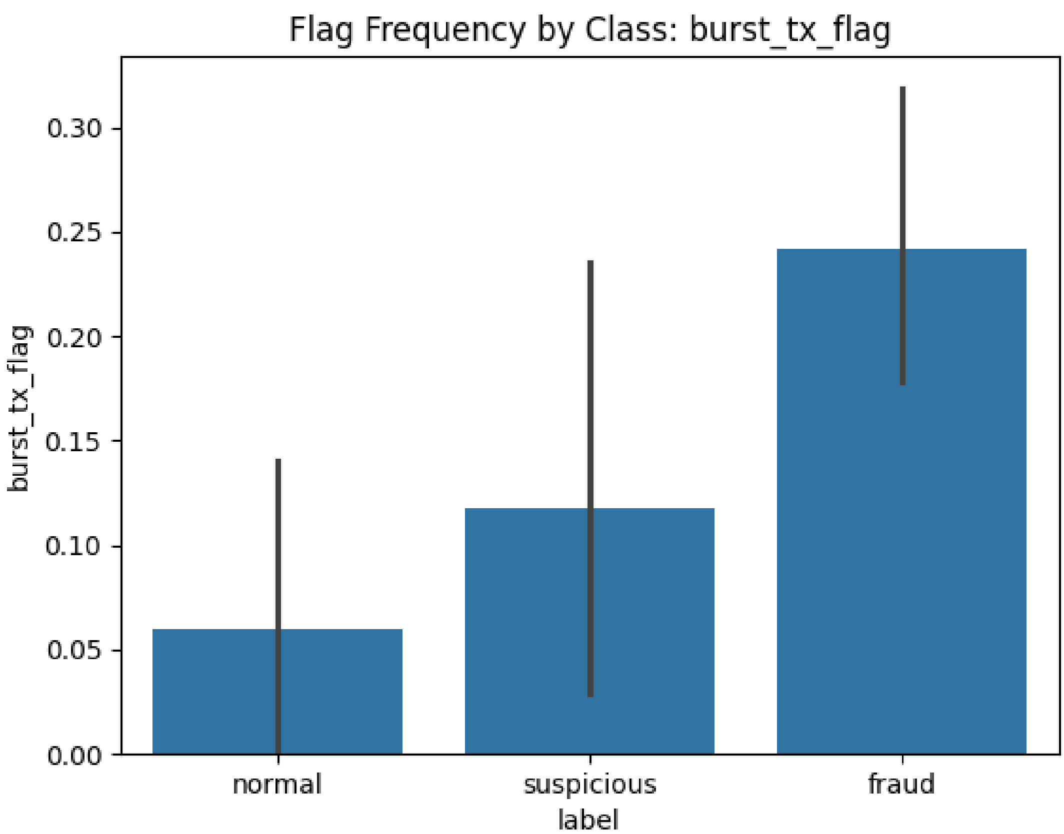

Figure 7.3.1a2: *burst_tx_flag_barplot.png*

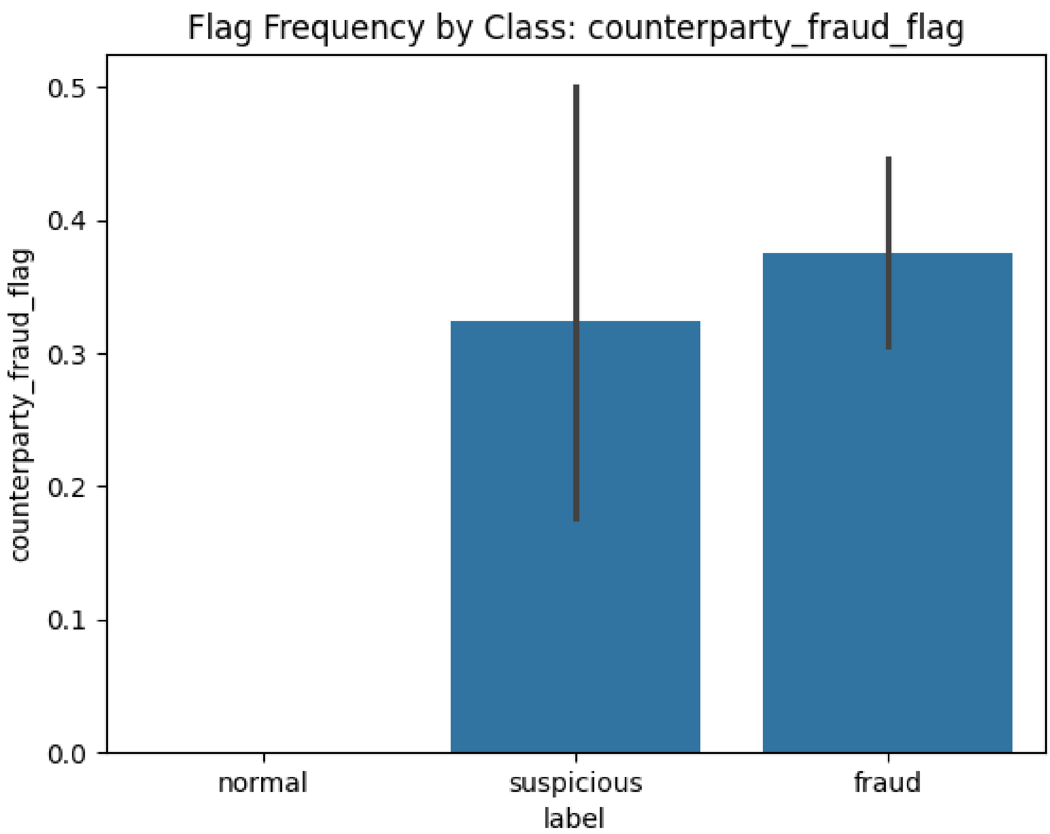

Figure 7.3.1a3: *counterparty_fraud_flag_barplot.png*

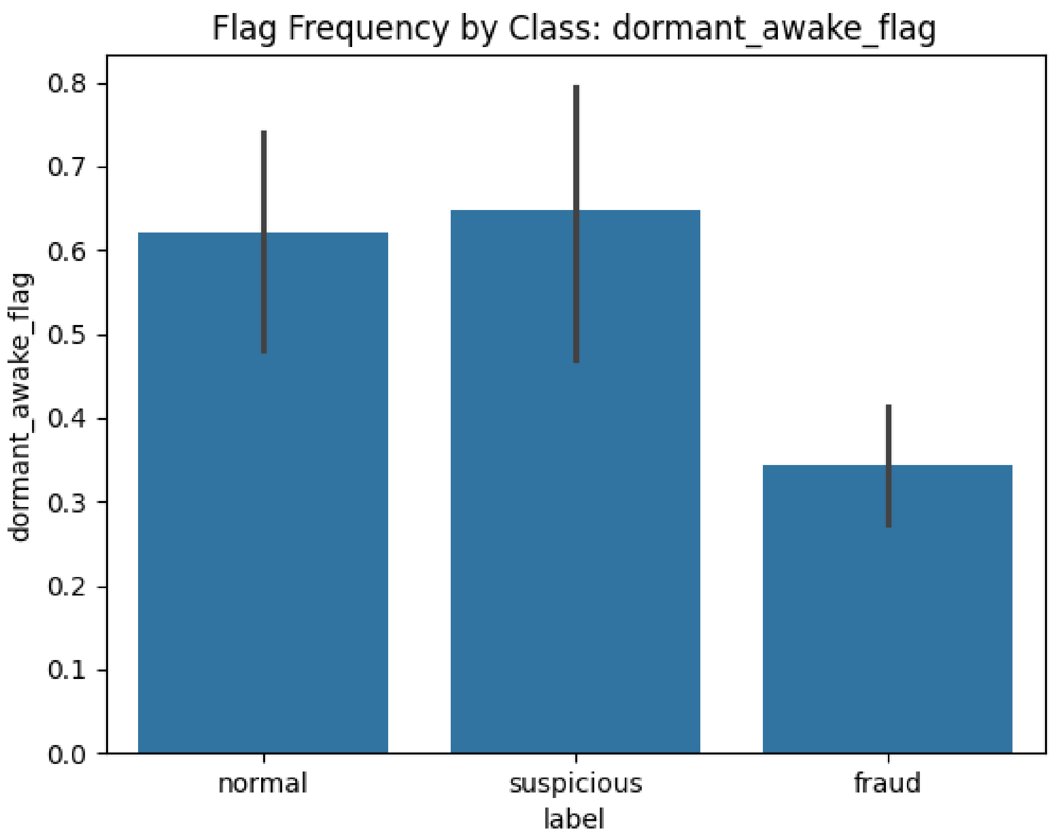

Figure 7.3.1a4: *dormant_awake_flag_barplot.png*

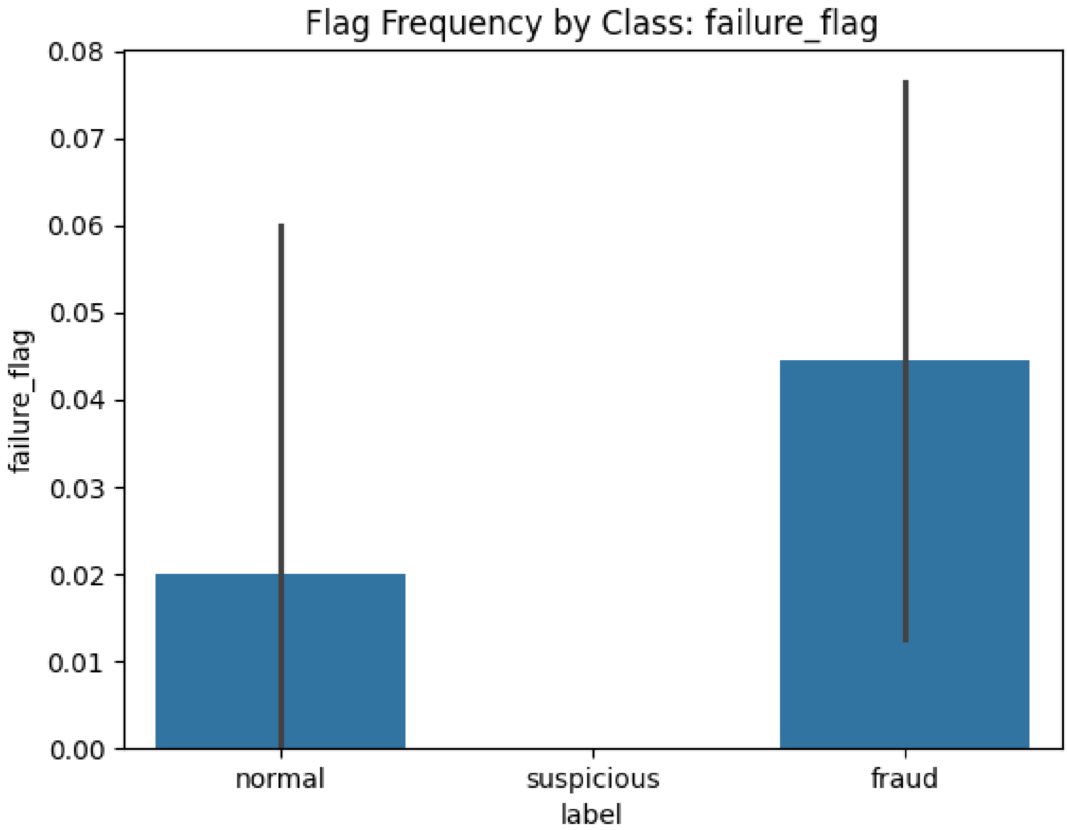

Figure 7.3.1a5: *failure_flag_barplot.png*

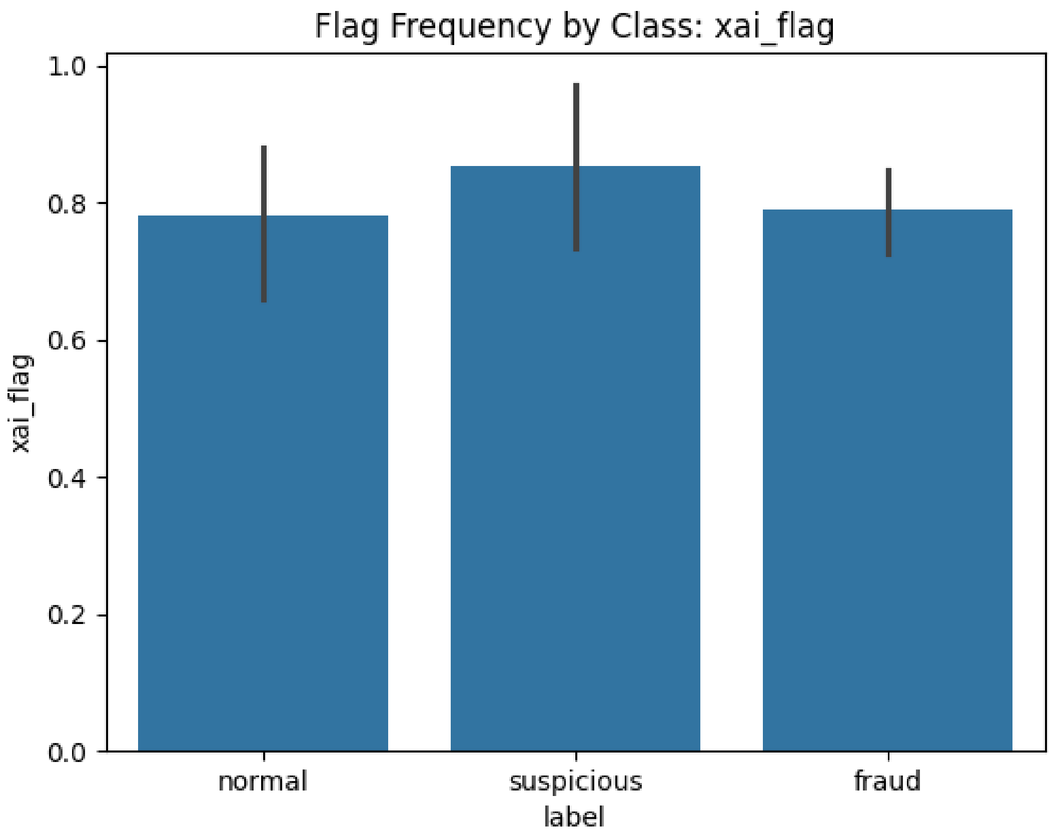

Figure 7.3.1a6*: xai_flag_barplot.png*

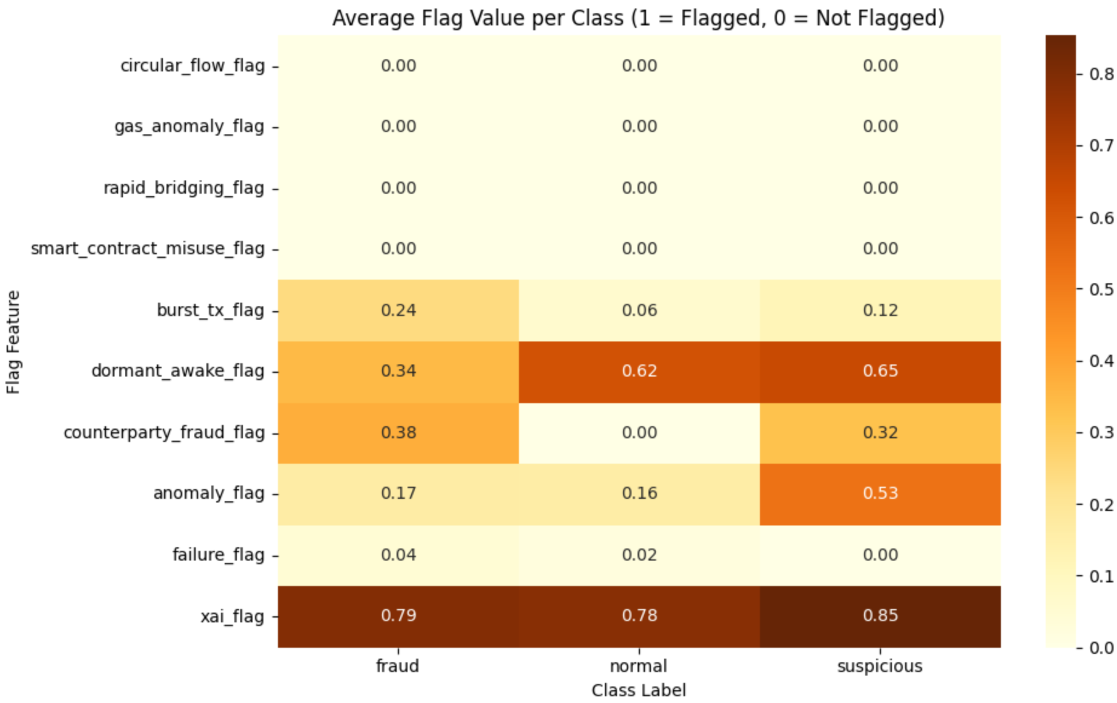

Figure 7.3.1a: *Risk Flag Heatmap*

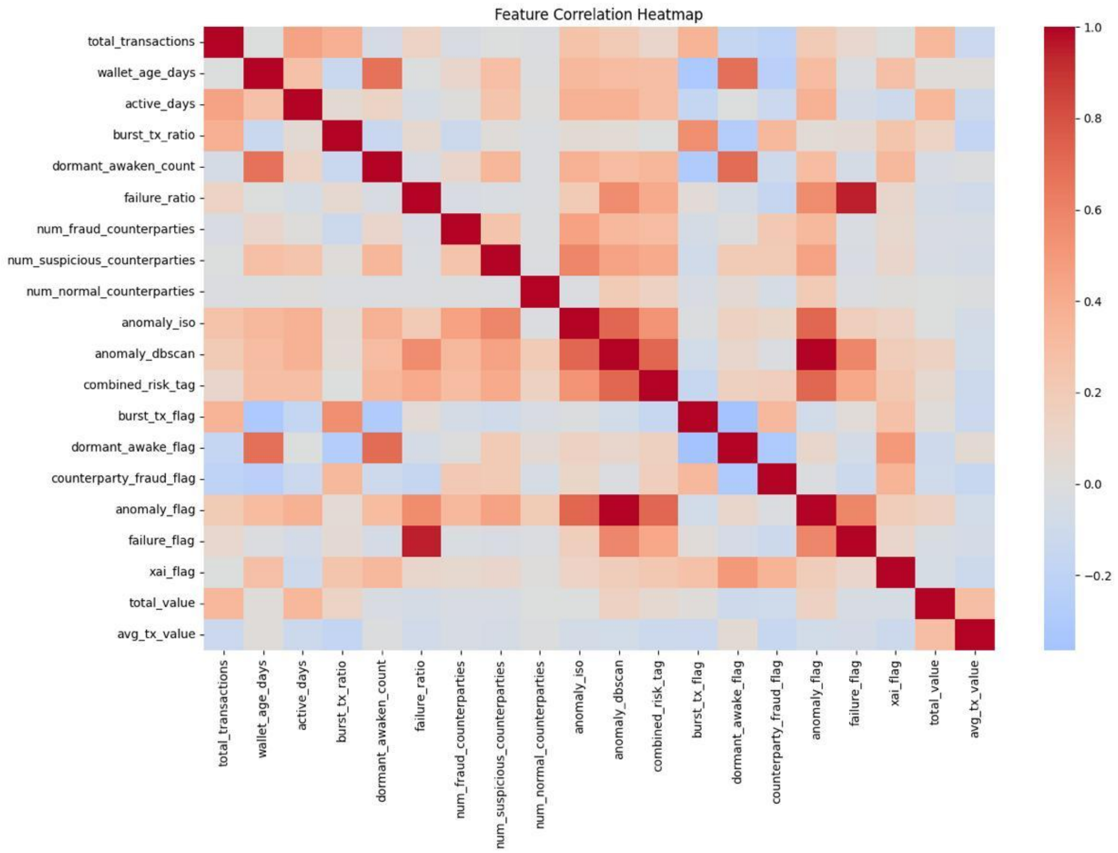

Figure 7.3.1b: Correlation Heatmap *(correlation_heatmap.png)*

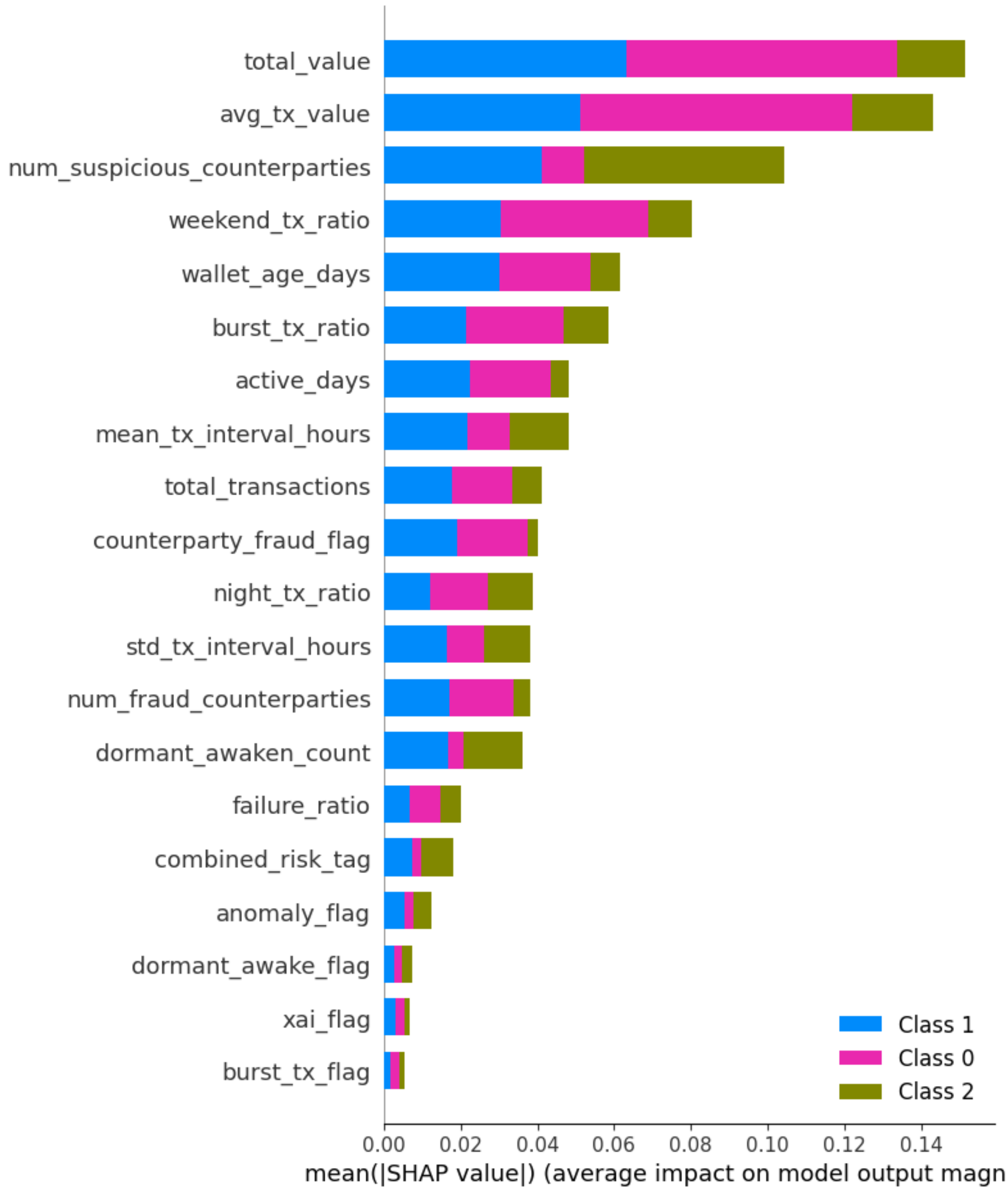

***Figure 7.3.2**: SHAP summary bar plot (RF)*

Table 7.1. Symbolic Flag Summary

Flag	Fraud	Normal	Suspicious
circular_flow_flag	0.00	0.00	0.00
gas_anomaly_flag	0.00	0.00	0.00
rapid_bridging_flag	0.00	0.00	0.00
smart_contract_misuse_flag	0.00	0.00	0.00
burst_tx_flag	0.24	0.06	0.12
dormant_awake_flag	0.34	0.62	0.65
counterparty_fraud_flag	0.38	0.00	0.32
anomaly_flag	0.17	0.16	0.53
failure_flag	0.04	0.02	0.00
xai_flag	0.79	0.78	0.85

Table 7.2. Feature Importances:

Rank	Layer	Feature	Importance Score
1	L0	total_value	0.149
2	L0	avg_tx_value	0.1411
3	L2	num_suspicious_counterparties	0.0898
4	L1	weekend_tx_ratio	0.0721
5	L0	wallet_age_days	0.0648
6	L1	burst_tx_ratio	0.064
7	L1	mean_tx_interval_hours	0.0602
8	L0	total_transactions	0.0575
9	L1	std_tx_interval_hours	0.0538
10	L0	active_days	0.0528
11	L1	night_tx_ratio	0.0434
12	L1	dormant_awaken_count	0.0368
13	L1	failure_ratio	0.0275
14	L2	num_fraud_counterparties	0.0226
15	L2	counterparty_fraud_flag	0.0156
16	L3	combined_risk_tag	0.0123
17	L3	anomaly_flag	0.0094
18	L1	dormant_awake_flag	0.0079
19	L3	xai_flag	0.0048
20	L2	num_normal_counterparties	0.0048
21	L3-XAI	burst_tx_flag	0.0038
22	L3	anomaly_iso	0.0037
23	L3-XAI	failure_flag	0.0023
24–30	L2–L3	*Other flags*	0

Table 7.3. Random Forest Classification Report:

Class	**Precision**	**Recall**	**F1-Score**
Normal (0)	0.90	0.90	0.90
Fraud (1)	0.86	0.94	0.90
Susp (2)	0.75	0.43	0.55
Accuracy	**0.86**		
Macro F1	**0.78**		

Table 7.4. XGBoost Classification Report:

Class	**Precision**	**Recall**	**F1-Score**
Normal (0)	0.77	1.00	0.87
Fraud (1)	0.87	0.84	0.86
Susp (2)	0.60	0.43	0.50
Accuracy	0.82		
Macro F1	0.74		

Table 7.5. GNN Classification Report:

Class	Precision	Recall	F1-Score
Normal (0)	0.50	0.50	0.50
Fraud (1)	0.67	0.91	0.77
Susp (2)	0.50	0.14	0.22
Accuracy	**0.62**		
Macro F1	**0.50**		

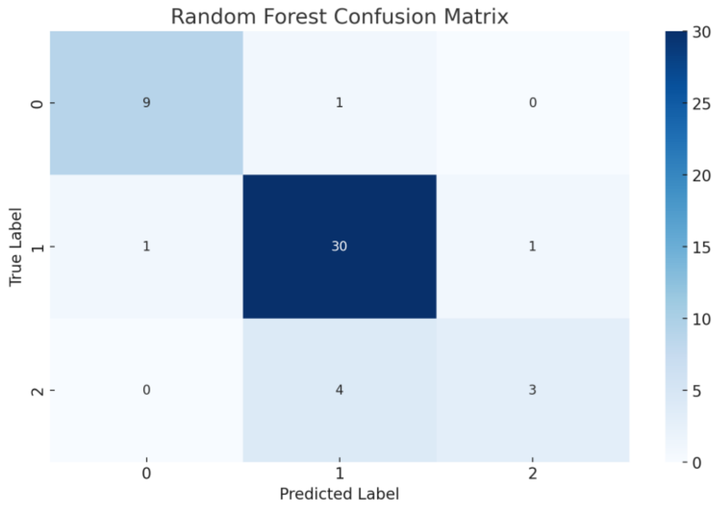

Figure 7.4.1: *Confusion matrix for RF*

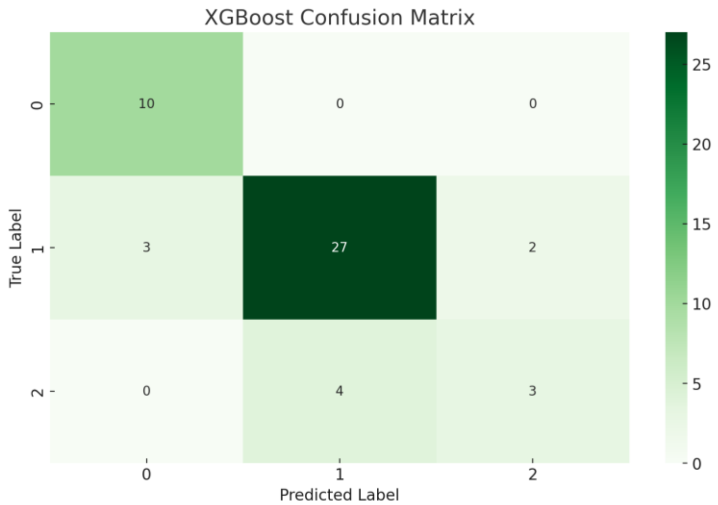

Figure 7.4.2: *Confusion matrix for XGB*

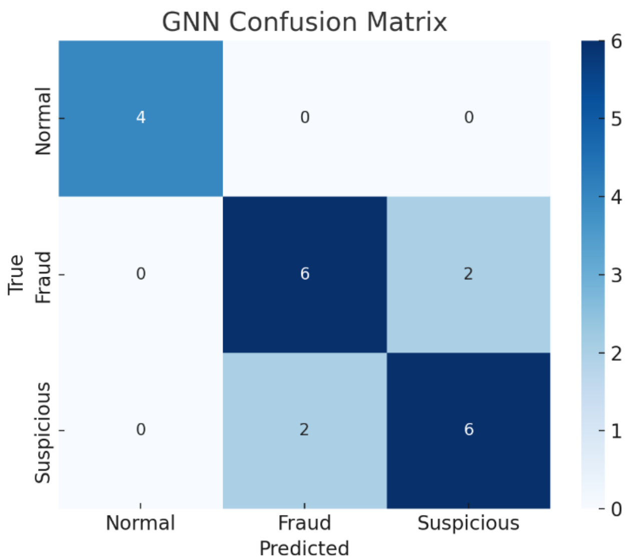

Figure 7.5.1: *Confusion matrix for GNN*

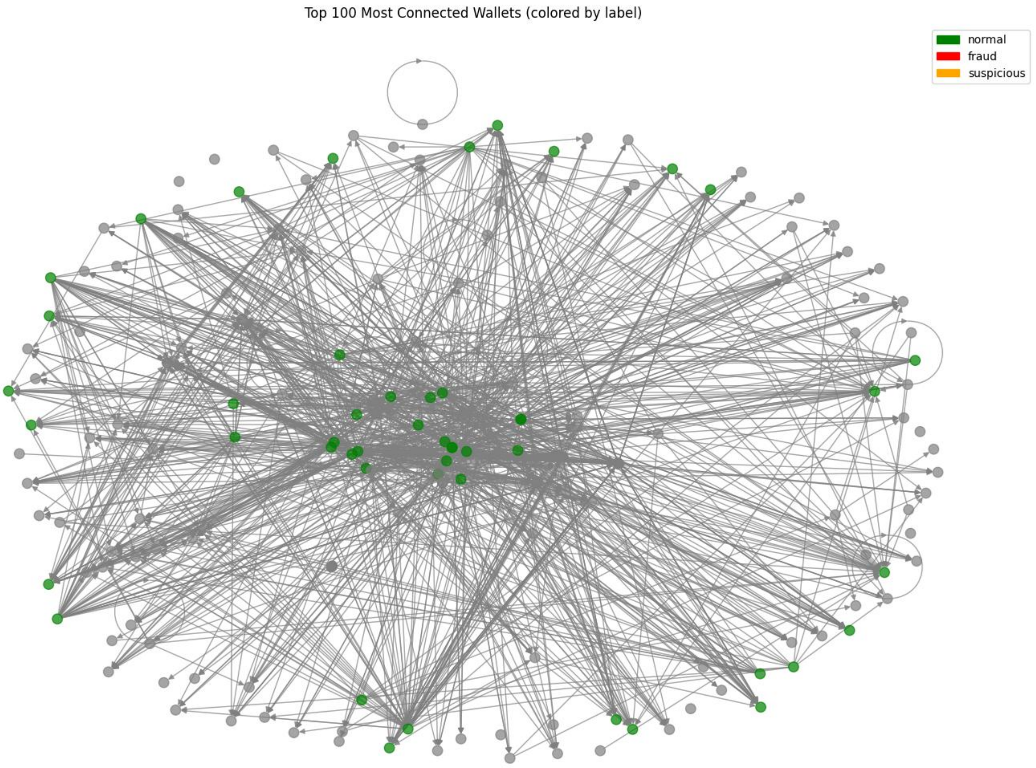

Figure 7.5.1: *Subgraph visualization – top connected fraud wallets*

Table 7.6. Node & Edge Summary report:

Node Label Distribution:

label_str	
fraud	155
normal	50
suspicious	34

Name: count, dtype: int64

Feature Statistics (mean/std/min/max):

	mean	std	min	max
total_transactions	1.290682e+03	2.949890e+03	1.0000	1.000000e+04
wallet_age_days	6.819498e+02	7.957678e+02	1.0000	3.593000e+03
active_days	7.693305e+01	1.594889e+02	1.0000	1.282000e+03
burst_tx_ratio	6.184312e-01	3.105239e-01	0.0000	1.000000e+00
dormant_awaken_count	2.263598e+00	3.239977e+00	0.0000	1.600000e+01
failure_ratio	4.470080e-02	1.676733e-01	0.0000	1.000000e+00
mean_tx_interval_hours	5.526327e+02	1.311278e+03	0.0001	9.256000e+03
std_tx_interval_hours	1.316442e+03	2.404073e+03	0.0100	1.552868e+04
weekend_tx_ratio	2.850682e-01	2.442168e-01	0.0000	1.000000e+00
night_tx_ratio	2.558985e-01	2.163817e-01	0.0000	1.000000e+00
num_fraud_counterparties	6.426778e+00	4.649724e+01	0.0000	6.040000e+02
num_suspicious_counterparties	3.054393e+00	1.674127e+01	0.0000	1.650000e+02
num_normal_counterparties	4.602510e-02	5.887379e-01	0.0000	9.000000e+00
anomaly_iso	5.020921e-02	2.188347e-01	0.0000	1.000000e+00
combined_risk_tag	2.594142e-01	4.392331e-01	0.0000	1.000000e+00
total_value	4.113390e+22	1.719183e+23	0.0000	1.924566e+24
avg_tx_value	2.966159e+20	8.895888e+20	0.0000	8.583729e+21

Edge List Summary:
num_edges: 77768
num_unique_sources: 52995
num_unique_targets: 5862
min_weight: 0.0
max_weight: 1000000.0
mean_weight: 576523.942868532
median_weight: 1000000.0
num_zero_weight: 32919
num_nonzero_weight: 44849

Degree Distribution (summary):

	out_degree	in_degree
count	58508.000000	58508.000000
mean	1.329186	1.329186
std	5.521672	65.035595
min	0.000000	0.000000
25%	1.000000	0.000000
50%	1.000000	0.000000
75%	1.000000	0.000000
max	903.000000	6844.000000

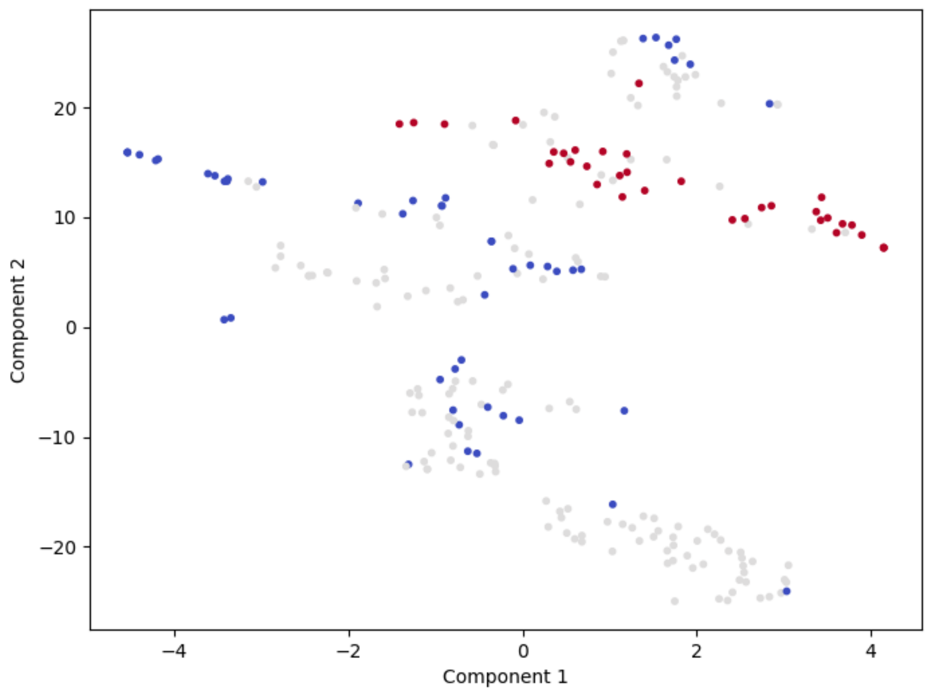

Figure 7.5.2: *tSNE 2D Visualization of Node Embeddings*

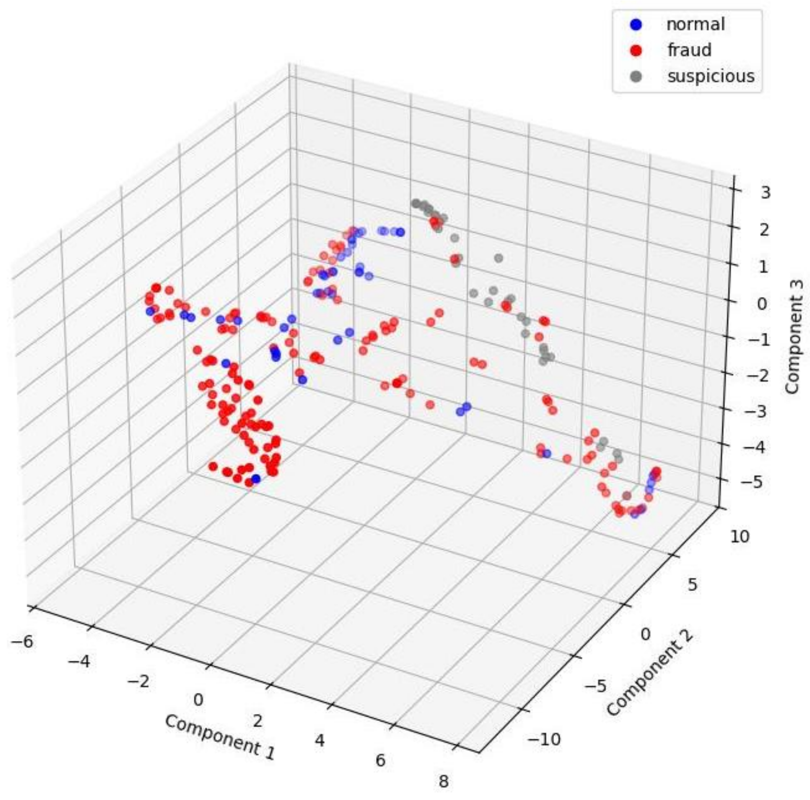

Figure 7.5.3: *tSNE 3D Visualization of Node Embeddings*

Appendix C: Limitations Observed

While the modular fraud detection framework demonstrated robust performance and interpretability, several limitations emerged during experimentation and deployment:

- **Feature Bias**: The models displayed dependence on high-volume or high-frequency features, potentially overlooking low-volume yet sophisticated fraud strategies designed to mimic benign behaviour.
- **Label Imbalance**: The dataset exhibited a skewed distribution, with underrepresentation of the "suspicious" class. This impaired the models' ability to learn nuanced boundaries between ambiguous behaviour and confirmed fraud.
- **Anomaly–Privacy Ambiguity**: Some anomaly signals such as mixer usage, burst transfers, or dormant awakenings—risked misclassification of privacy-preserving activity as fraudulent, particularly in edge cases.
- **Graph Sparsity**: The GCN implementation operated on a restricted subgraph of 239 labelled nodes out of ~58,000 due to strict completeness filters. This limited the graph's capacity to capture relational learning and hindered generalization.
- **GCN Performance Constraints**: Despite achieving high recall for fraud, the GCN model suffered from low precision and overall accuracy (~62%), attributed to class imbalance, low edge density, and absence of contextual edge features (e.g., timestamps, token types).
- **Temporal Blind Spots**: The current models lack dynamic sequence modelling, missing key temporal fraud indicators such as fund-hopping delays, flash loan cycles, or orchestrated liquidation events.
- **Graph Fragmentation**: Network visualizations revealed that many wallet subgraphs were disconnected or weakly linked, reducing the effectiveness of message propagation and limiting the discovery of fraud rings or collaborative behaviour.

Appendix D: Future Work

To mitigate these limitations and evolve the framework toward production-grade blockchain AML systems, the following enhancements are recommended:

- **Label Enrichment & Rebalancing**: Use techniques like SMOTE (Synthetic Minority Over-sampling), weak supervision, or adversarial data generation to increase representation of suspicious wallets and sharpen model boundaries.
- **Temporal Feature Engineering**: Integrate dynamic features such as inter-transfer delays, looping frequencies, transaction session gaps, and flash loan footprints to improve detection of coordinated, time-based attacks.
- **Graph Expansion via Semi-Supervised Learning**: Apply label propagation, bootstrapping, or self-training GNNs to incorporate unlabelled or partially observed wallets into the graph learning pipeline without relying solely on ground-truth labels.
- **GNN Architectural Exploration**: Benchmark advanced models such as:
- **Graph Attention Networks (GAT)** for node importance weighting.
- **GraphSAGE** for inductive learning on unseen wallets.
- **Temporal GCNs** for modelling fraud evolution over time.
- **Cross-Chain & Layer-2 Scaling**: Extend Noir's scope to support multi-chain behaviour fingerprinting across bridges, rollups, and wrapped assets to detect inter-chain laundering.
- **Causal & Counterfactual Explainability**: Add reasoning modules that simulate "what-if" scenarios or counterfactual paths to enhance human trust, especially for regulators seeking traceable audit logs.
- **Real-Time Simulation Environment**: Build a pipeline to ingest live blockchain activity (e.g., from mempool, testnets) and test Noir's response to adversarial wallets, suspicious contract launches, or flash attack sequences under streaming conditions.

Appendix E: Data Sources

The datasets used in this thesis are sourced entirely from public blockchain infrastructure, community-tagged repositories, and open-access APIs. This approach ensures reproducibility, transparency, and accessibility, key to building trust in a modular fraud detection framework.

E.1 Blockchain Transaction APIs

Source	***Purpose***	***Access***	***Relevance***
Etherscan API	Ethereum mainnet transactions	etherscan.io/apis	Core L1 transaction data source; used to track wallet activity, gas, and timing behaviour.
Arbiscan API	Arbitrum Layer-2 transactions	arbiscan.io/apis	Captures L2 activity often excluded from L1-only analytics.
Optimism Explorer API	Optimism L2 transactions	optimistic.etherscan.io	Adds L2 visibility, important for detecting cross-chain flow manipulation.
BaseScan API *(Optional)*	Coinbase-backed L2 data	basescan.org/apis	Expands scope to emerging ecosystems. Used for comparative robustness testing.
Flipside Crypto	Curated SQL-based blockchain datasets	flipsidecrypto.xyz	Enables sampling from high-activity or suspicious wallets. Also supports behaviour aggregation.
Dune Analytics	Community dashboards and custom queries	dune.com	Used to extract incident-driven or historical fraud cases (e.g., post-hack tracing).

E.2 Labelled Fraud / AML Datasets

Because there is no centralized AML labelling mechanism on public blockchains, this project uses a **proxy labelling strategy** that combines multiple public and semi-trusted sources:

- **Government-Sanctioned Lists**: Wallets listed by regulatory bodies such as the U.S. Treasury's **OFAC SDN list** provide high-confidence fraud indicators, although they cover a relatively narrow subset of illicit actors.
- **Crowdsourced Scam Repositories**: Platforms like **ScamSniffer**, **MetaMask's scam archive**, and **Chainabuse** collect and publish scam-related wallet addresses based on user reports and community moderation. These are useful for pattern recognition and feature validation.
- **Open Community Tagging Platforms**: Tools like **Etherscan** maintain real-time wallet labels (e.g., "Phishing", "Scam", "Fake_Token") which help provide semi-supervised learning signals across Ethereum and L2 ecosystems.

Source	***Description***	***Access***	***Use Case***
OFAC Sanctioned Wallets	U.S. Treasury SDN-listed addresses	OFAC SDN List	Used for high-confidence fraud labeling.
Etherscan Labels	Tags like Phishing, Scam, Fake_PhishingXXX	etherscan.io	Provides semi-supervised wallet-level labels.
ScamSniffer GitHub Dataset	Real-time scam wallet list	ScamSniffer GitHub	Structured and updated JSON/CSV; useful for real-world fraud detection benchmarking.

Chainabuse	Community-reported scams (no API)	chainabuse.com	Provides addresses tied to real-world scam reports. Data is scraped and documented.
MetaMask Phishing Archive	Historic phishing and scam records	MetaMask GitHub	Used for feature augmentation and fraud frequency validation.

E.3 Live Testing / Production Sim Data

To evaluate generalizability under realistic conditions, a live testing pipeline was deployed. This includes:

- **High-value wallet tracking** via Etherscan leaderboard
- **Cross-chain movement detection** using Flipside SQL (e.g., mixer-bridge combinations)
- **Emerging wallets** (e.g., new contract deployers) monitored for suspicious flows

These live scenarios simulate production-grade behaviour scoring and anomaly detection in real time.

Appendix F: Noir Framework Execution Workflow

This appendix outlines the full technical workflow executed for the Noir Framework, capturing the modular scripts, their order of execution, and the pipeline stages they represent. It serves to validate the implementation effort and clarify the end-to-end data and model handling strategy.

F.1 Overview of Workflow Phases

The Noir Framework is structured as a modular, multi-phase system with the following core phases:

Phase	Objective	Main Scripts Used
Phase 1	Data collection, cleaning, feature engineering	data_fetch_*.py, merge_transaction_sets.py, build_features_l0_aggregate.py, etc.
Phase 2	Model training and risk classification	prepare_features_for_training.py, train_supervised_models.py
Phase 3	Interpretability and explainability	build_features_l3_xai_tags.py, shap_explain_v2.py, analyze_feature_flags.py
Phase 4	GNN-based laundering flow modeling (prototype)	build_graph_dataset.py, prep_gnn_input.py, train_gnn_model.py
Phase 5	Experimental evaluation (planned but not implemented)	evaluate_predictions_v2.py, prepare_training_rerun.py

F.2 Script-by-Script Breakdown

Pipeline Stage 1: Data Acquisition and Preparation

- data_fetch_fraud.py / data_fetch_mixers.py / data_fetch_normal.py: Collect raw transaction data.
- merge_transaction_sets.py: Merge all categories (fraud, normal, mixer).
- validate_merged_data.py: Sanity checks and validations.

Pipeline Stage 2: Feature Engineering

- build_features_l0_aggregate.py: Transaction count, volume, wallet age.
- build_features_l1_behavior.py: Time-based behaviours and burst/dormancy.
- build_features_l2_riskflags.py: Interaction with known fraud/mixers.
- build_features_l3_metaai.py: Isolation Forest, DBSCAN anomaly flags.
- build_features_l3_xai_tags.py: Human-readable reason codes + tags.
- merge_final_features.py: Merge L0–L3 into master feature table.

Pipeline Stage 3: Model Training and Evaluation

- prepare_features_for_training.py: Final feature cleaning and splits.
- train_supervised_models.py: Random Forest and XGBoost model training.

Pipeline Stage 4: Graph Neural Network Modelling

- build_graph_dataset.py: Transform wallet data into node-edge graph.
- prep_gnn_input.py: Clean, encode and structure for PyTorch GCN.
- train_gnn_model.py: Train GCN and evaluate results.
- analyse_predictions.py: Cluster analysis and visual inspection.

Pipeline Stage 5: Analysis and Reporting

- analyze_feature_stats.py, transaction_stats_summary.py: Feature and flow insights.
- analyze_feature_flags.py: Breakdown of risk flags across wallet types.
- shap_explain_v2.py: Model explainability visualizations.

F.3 Execution Summary

The pipeline was built in a reproducible and modular way. Each script could be independently rerun, facilitating debugging and experimentation. The modular architecture supports future integration of causal simulation (Phase 4) and generalization (Phase 5).

Appendix G: Technical Stack

The project employs an open-source and modular stack spanning data ingestion, ML pipelines, explainability, and graph-based visualization.

Component	Tools & Libraries	Purpose
Data Collection	requests, etherscan-python, web3.py, Dune SDK, Selenium	API calls and scraping from chains and tagging sites
Data Preparation	pandas, numpy, datetime, sqlite3, json	Data parsing, aggregation, time normalization
Feature Engineering	pandas, numpy, custom scripts	Extract wallet behaviours (e.g., dormant triggers, mixers, hops)
Unsupervised Detection	scikit-learn (Isolation Forest, DBSCAN)	Discover unknown patterns from unlabelled data
Supervised Modeling	scikit-learn, xgboost, imblearn	Classify fraud using known tags from OFAC, Etherscan
Explainability	shap (planned), rule-based templates, feature importances	Post-prediction reasoning and risk code generation
Visualization	matplotlib, seaborn, plotly, networkx, pyvis	Visual analytics, flows, correlation views
Deployment/Live Testing	joblib, ETL scripts	Evaluate models on live L2 streams
Experiment Tracking	Jupyter, structured logs	Track validation and model evolution across iterations

Appendix H: Deliverables & Timeline

The final output of this thesis will include both code artifacts and analytical documentation that demonstrate the practical implementation and theoretical insights of the proposed fraud detection system.

1. **End-to-End ML Pipeline**:
   - Modular scripts for data processing, feature engineering, model training, evaluation, and live testing.
2. **Trained Models and Evaluation Logs**:

   - Pickled models, performance metrics (confusion matrices, classification reports), and validation results.
3. **Fraud-Flagged Wallet Output**:
   - CSV files containing predictions, anomaly probabilities, and interpretability tags (e.g., reason codes).
4. **Visualization and Explainability Assets**:
   - SHAP plots, dashboard-ready charts, and feature importance visuals to support auditability.
5. **Thesis Report**:
   - Full academic write-up with figures, literature review, case studies, evaluation results, and discussion of failure modes or improvement areas.
6. **(Optional) Modular Design Document**:
   - Architecture diagram, layer-by-layer logic, and roadmap for future extensions of the fraud detection framework.

Timeline and Planning

Milestone	Date	Deliverables & Activities
Data Collection	March 2025	Historical + live transaction data from Ethereum and Layer-2s collected.
Feature Engineering	March 2025	Wallet-level features engineered (dormant triggers, bursts, etc.); will be refined in June.
GitHub Repository Setup	June 18, 2025	Repo initialized at Noir Framework with code structure and documentation.
TOC + Draft 1 (Method + Contributions)	June 24, 2025	Submit contributions section, system architecture (Section 5), and planned evaluation logic.
Draft 2 (Detection Output + Evaluation Plan)	July 3, 2025	Share fraud prediction outputs (Detection Layer), explain evaluation approach (metrics, thresholds, tags).
Demo to Supervisor	July 8, 2025	Live walkthrough of working fraud detection pipeline with SHAP/rule-based explainability.
Full Draft Submission	~~July 17, 2025~~ July 23, 2025	Complete thesis draft (Sections 5–14) including evaluation results, charts, reason codes, discussion and conclusion.
Revised Draft (Post Feedback)	Aug 7, 2025	Integrate supervisor feedback, polish visuals, and finalize formatting.
Sync Touchpoints	Aug 12 & Aug 19, 2025	Interim check-ins to align on any final edits or clarifications.
Final Submission Window	*Aug 27–29, 2025*	Submit final PDF, GitHub code artifacts, and presentation-ready assets.

This timeline ensures that all deliverables, including the refined feature engineering and live testing modules, will be completed and ready for review in July and final submission by August 29th, 2024.

Appendix I: Glossary

Blockchain & Wallet Terms	
Blockchain	A decentralized digital ledger where transactions are recorded across a distributed network.
Wallet	A blockchain address used to send, receive, or store cryptocurrency. It can belong to a user, bot, or smart contract.
Smart Contract	Self-executing code deployed on a blockchain, triggered when predefined conditions are met.
Bridge	A protocol enabling token or data transfers between blockchains (e.g., Ethereum ↔ Arbitrum).
Mixer	A service that obscures transaction origins by blending coins, often used for privacy or laundering.
Dormant Wallet	An inactive wallet that suddenly becomes active — potentially signalling fraud.
Circular Flow	A transaction loop where funds return to the origin, often to obscure money trails.
Layer-2 (L2)	A secondary protocol built on top of a main blockchain (e.g., Arbitrum on Ethereum) to improve speed and reduce costs.
Ethereum (ETH)	A widely used blockchain platform supporting smart contracts and decentralized apps.
Arbitrum / Optimism	L2 rollups for Ethereum that scale transaction throughput.
Machine Learning & AI Terms	
Supervised Learning	Training models on labelled data (e.g., known fraud vs. normal).
Unsupervised Learning	Detecting patterns in unlabelled data, often used for anomaly detection.
Random Forest (RF)	An ensemble of decision trees that improves classification accuracy.
XGBoost (XGB)	A fast and accurate gradient-boosted model for tabular classification.
Isolation Forest	An unsupervised model that identifies anomalies by isolating data points.
DBSCAN	A clustering algorithm that detects dense behavioural clusters and outliers.
Explainability & Interpretability	
SHAP (SHapley Additive Explanations)	A method to quantify each feature's impact on a model's output.
XAI (Explainable AI)	A set of methods that make machine learning decisions transparent and auditable.
Reason Codes	Human-readable tags (e.g., burst_tx + fraud_link) that explain why a wallet was flagged.
Noir Framework Feature Layers	
L0 (Aggregate)	Basic stats (e.g., total tx count, average value, wallet age).
L1 (Behavioural)	Activity patterns (e.g., bursts, dormancy, night/weekend usage).
L2 (Risk Flags)	Rule-based signals (e.g., mixer use, fraud links, circular flows).

L3 (Meta-AI)	Unsupervised model outputs like anomaly scores or clusters.
L3 (XAI Tags)	Binary tags indicating risky behaviour + reason codes for interpretability.
Graph & Network Terms	
Graph Neural Network (GNN)	A model that learns from structured data like wallet networks.
GCN (Graph Convolutional Network)	A specific GNN variant using neighbouring node information.
Node	A point in a graph (e.g., a wallet).
Edge	A connection between nodes, typically a transaction.
Wallet Cluster	A group of interconnected wallets sharing similar behaviour.
Compliance & Fraud Detection	
AML (Anti-Money Laundering)	Practices to detect and prevent illicit financial activity.
Fraudulent Wallet	A wallet involved in scams, phishing, or laundering (per labels or detection).
Suspicious Wallet	A wallet showing risky behaviour but not conclusively flagged as fraud.
OFAC (Office of Foreign Assets Control)	U.S. authority managing sanction lists, including crypto wallets.
Anomaly Detection	ML techniques used to identify behaviour deviating from the norm.
Fraud Link Count	A derived feature counting interactions with known fraudulent wallets.
Transaction Burst	A sudden spike in transaction frequency, often flagged as anomalous.